

Contents

1	High Level Synthesis	5
1.1	UG902 ???	5
2	References	5
3	Introduction	5
3.1	Programming Model	6
4	FPGA Architecture	7
4.1	Look Up Table (LUT)	8
4.2	Flip-Flop	8
4.3	DSP Block	8
4.4	Storage Elements	9
4.4.1	BRAM	9
4.4.2	URAM	9
4.4.3	LUT	9
4.4.4	Shift Register	10
5	FPGA Parallelism Versus Processor Architectures	10
5.1	Program Execution on a Processor	10
5.2	Program Execution on an FPGA	10
5.2.1	Scheduling	11
5.2.2	Pipelining	11
5.2.3	Dataflow	11
6	Basic Concepts of Hardware Design	12
6.1	Clock Frequency	12
6.1.1	Instruction execution	12
6.1.2	Power Consumption	13
6.2	Latency and Pipelining	14
6.2.1	Important points	14
6.3	Throughput	15
6.4	Memory Architecture and Layout	15
6.4.1	Registers	16
6.4.2	Shift Register	16
6.4.3	FIFO	17
6.4.4	BRAM	17
7	Vivado High-Level Synthesis	17
7.1	Operations	17
7.1.1	Optimizations by vivado HLS	18
7.2	Conditional Statements	18
7.3	Loops	19
7.3.1	Optimizations by vivado HLS	19
7.4	Functions	19
7.5	Dynamic Memory Allocation	19
7.6	Pointers	21

8	Software Verification and Vivado HLS	21
8.1	Software Test Bench	21
8.2	Code Coverage	22
8.3	Uninitialized Variables	22
8.4	Out-of-Bounds Memory Access	23
8.5	Co-Simulation	23
8.6	When C/C++ Verification Is Not Possible	24
9	Design Example: Application Running on a Zynq-7000 SoC	24
9.1	Analyzing and partitioning the processor code	24
9.2	Compiling the program in Vivado HLS	25
9.3	Composing the system in Vivado IP integrator	25
9.4	Connecting processor code and FPGA fabric functions	26
10	Verification of a Complete Application	26
10.1	Standalone compute systems	26
10.1.1	Module Verification	26
10.1.2	Connectivity Verification	27
10.1.3	Application Verification	28
10.1.4	Device Validation	28
10.2	Processor-based systems	28
10.2.1	Hardware in the Loop (HIL) Verification	28
10.2.2	Virtual Platform(VP) Verification	29
10.2.3	Device Validation	29
11	Design Principles for Software Programmers	29
11.1	Introduction	29
11.2	Computer Architecture	29
11.3	Three Paradigms for Programming FPGAs	30
11.3.1	Producer-Consumer Paradigm	31
11.3.2	Streaming Data Paradigm "Dataflow"	31
11.3.3	Pipelining Paradigm	31
11.4	Combining the Three Paradigms	32
11.5	Conclusion - A Prescription for performance	33
12	Introduction to Vitis/Vivado HLS	34
12.1	High-Level Synthesis Benefits	34
12.2	High-Level Synthesis Basics	35
13	Vitis/Vivado HLS Process Overview	35
14	Creating project with Vitis HLS	37
14.1	Coding C/C++ Functions	38
14.1.1	Coding Style	38
14.1.2	C/C++ Language Support	38
14.1.3	Libraries in Vitis HLS	38
14.2	Specifying the Clock Frequency	39
15	C simulation verification	39

15.1 Writing a Test Bench	39
16 C Synthesis	40
17 Analyzing the Results of Synthesis	40
17.1 Schedule Viewer	41
17.2 Function Call Graph Viewer	41
17.3 Dataflow Viewer	42
18 Optimizing the HLS Project	42
18.1 Creating Additional Solutions	43
18.2 Adding Pragmas and Directives	43
18.3 Using Directives in Scripts vs Pragmas in Code	43
18.3.1 Directives file (Tcl Script)	43
18.3.2 Source Code (Pragma)	43
18.3.3 Applying Directives to the Proper Scope	43
18.3.4 Applying Optimization Directives to Global Variables	44
18.3.5 Applying Optimization Directives to Class Objects	44
18.3.6 Applying Optimization Directives to Templates	44
19 C/RTL Co-Simulation in Vitis HLS	44
19.1 Automatically Verifying the RTL	44
19.1.1 Verification of DATAFLOW and DEPENDENCE	45
19.1.2 Unsupported Optimizations for Co-Simulation	46
19.2 Analyzing RTL Simulations	46
19.3 Cosim Deadlock Viewer	46
19.4 Debugging C/RTL Co-Simulation	47
20 Exporting the RTL Design	47
21 Vitis HLS Command Line Interface	47
21.1 Running Vitis HLS Interactively	47
21.2 Running Vitis HLS in Batch Mode	48
22 Designing Efficient Kernels	48
23 Vitis HLS Coding Styles	48
23.1 Unsupported C/C++ Constructs	48
23.1.1 System Calls	49
23.1.2 Dynamic Memory Usage	49
23.1.3 Pointer Limitations	49
23.1.4 Recursive Functions	50
23.2 Functions	50
23.2.1 Inlining Functions	50
23.2.2 C/C++ Builtin Functions	50
23.3 Loops	50
23.3.1 Rolled implementation	51
23.3.2 Full loop unroll	51
23.3.3 Partial loop unroll	51

23.3.4	Loop Pipelining	52
23.3.5	Nested Loop	52
23.3.6	Loop flattening	54
23.3.7	Loop Merging	54
23.3.8	Variable Loop Bounds	54
23.3.9	Loop Parallelism	55
23.3.10	Loop Dependencies	55
23.4	Arrays	55
23.4.1	Minimize the parallel array access	56
23.4.2	Array merging	56
23.4.3	Array partitioning	56
24	Defining Interfaces	57
25	Optimization Techniques in Vitis HLS	57
26	High-Level Synthesis Introduction	58
26.1	Lab 1: Creating a High-Level Synthesis Project	58
26.1.1	HLS Graphical User Interface (GUI)	58
26.1.2	Usual self check operation of test bench	59
26.1.3	C synthesis report	59
26.1.4	RTL co-simulation	59
26.2	Lab 2: Using the Tcl Command Interface	59
26.2.1	Important points to note	60
26.3	Lab 3: Using Solutions for Design Optimization	60
27	C Validation	61
27.1	Lab 1: C Validation and Debug	61
27.1.1	Good practices for writing a testbench	61
27.2	Lab 2: C Validation with ANSI C Arbitrary Precision Types	61
27.3	Lab 3: C Validation with C++ Arbitrary Precision Types	62
28	Interface synthesis	62
28.1	Lab 1: Block-Level I/O Protocols	62
28.1.1	Important points to be noted	62
28.1.2	Block level I/O Protocol	62
28.1.3	Types of the block-level interface protocol	63
28.2	Lab 2: Port I/O Protocols	63
28.2.1	Important points to be noted	63
28.3	Lab 3: Implementing Arrays as RTL Interfaces	64
28.3.1	Important points to be noted	64
28.4	Lab 4: Implementing AXI4 Interfaces	65
29	Arbitrary Precision Types	65
29.1	Lab 1: Arbitrary Precision	65
29.2	Lab 2: Arbitrary Precision	66
30	Design Analysis	66
30.1	Lab 1: Design Optimization	66

30.1.1 Important points to be noted	67
31 Design Optimization	68
31.1 Lab 1: Optimizing a Matrix Multiplier	68
31.1.1 Aim	68
31.2 Lab 2: C Code Optimized for I/O Accesses	69
32 RTL Verification	70
32.1 Lab 1: RTL Verification and the C Test Bench	70
32.2 Lab 2: Viewing Trace Files in Vivado	70
32.3 Lab 3: Viewing Trace Files in ModelSim	71
33 Using HLS IP in IP Integrator	71
33.1 Lab 1: Integrate HLS IP with a Xilinx IP Block	71
34 Using HLS IP in a Zynq AP SoC Design	71
34.1 Lab 1: Implement Vivado HLS IP on a Zynq Device	71
34.2 Lab 2: Streaming Data Between the Zynq CPU and HLS Accelerator Blocks	72

1 High Level Synthesis

- High-Level Synthesis Prerequisites(UG998)
- Vitis High-Level Synthesis(UG1399)
- Vivado High-Level Synthesis Tutorial(UG871)
-

1.1 UG902 ???

2 References

- Vitis High-Level Synthesis User Guide (UG1399)
- Introduction to FPGA Design with Vivado High-Level Synthesis (UG998)
- Vivado Design Suite User Guide High-Level Synthesis (UG902)
- Vivado Design Suite Tutorial High-Level Synthesis Tutorial (UG871)
- Hardware Acceleration Labs Github
- Vitis HLS Migration Guide (UG1391)

3 Introduction

Software is the basis of all applications. Based on the performance and programmability constraints of the system, the software engineer is tasked with determining the best implementation platform to get a project to market. To accomplish this task, the software engineer is aided by both programming techniques and a variety of hardware processing platforms.

On the programming side, previous decades yielded advances in object-oriented programming for code reuse and parallel computing paradigms for boosting algorithm performance. The advancements in programming languages, frameworks, and tools allowed the software engineer to quickly prototype and test different approaches to solve a particular problem.

Although the interest in the parallel and concurrent execution of software programs is not new, the renewed and increased interest is aided by certain trends in processor and application-specific integrated circuit (ASIC) design. There are two crucial factors for getting more performance out of a software algorithm with the use of a custom-integrated circuit or an FPGA:

- Cost to fabricate the circuit
- Time to translate the algorithm into hardware

Despite advancements in fabrication process node technology that have yielded significant improvements in power consumption, computational throughput, and logic density, the cost to fabricate a custom-integrated circuit or ASIC for an application is still high. At each processing node, the cost of fabrication continues to increase to the point where this approach is only economically viable for applications that ship in the range of millions of units.

The second option is to use an FPGA, which addresses the cost issues inherent in ASIC fabrication. FPGAs allow the designer to create a custom circuit implementation of an algorithm using an off-the-shelf component composed of basic programmable logic elements. This platform offers the power consumption savings and performance benefits of smaller fabrication nodes without incurring the cost and complexity of an ASIC development effort. Similar to an ASIC, an algorithm implemented in an FPGA benefits from the inherent parallel nature of a custom circuit.

3.1 Programming Model

The programming model of a hardware platform is one of the driving factors behind its adoption. Software algorithms are typically captured in C/C++ or some other high-level language, which abstracts the details of the computing platform. These languages allow for quick iteration, incremental improvements, and code portability, which are critical to the software engineer.

With the recent paradigm shift in the design of standard and specialized processors, both types of processors stopped relying on clock frequency increases for program speedup and added more processing cores per chip. Multicore processors put program parallelization at the forefront of techniques used to boost software performance. The software engineer must now structure algorithms in a way that leads to efficient parallelization for performance. The techniques required in algorithm design use the same base elements of FPGA design. The main difference between an FPGA and a processor is the programming model.

Historically, the programming model of an FPGA was centered on register-transfer level (RTL) descriptions instead of C/C++. Although this model of design capture is completely compatible with ASIC design, it is analogous to assembly language programming in software engineering.

Both the initial and optimized versions of an application with an FPGA platform provide significant performance when compared against the same stages for both standard and specialized processors. RTL coding and an FPGA optimized application result in the highest performance implementation. However, the development time required to arrive at this implementation is beyond the scope of a typical software development effort. Therefore, FPGAs were traditionally used only for those

applications requiring a performance profile that could not be achieved by any other means, such as designs with multiple processors.

Recent technological advances by Xilinx remove the difference in programming models between a processor and an FPGA. Just as there are compilers from C and other high-level languages to different processor architectures, the Xilinx Vivado High-Level Synthesis (HLS) compiler provides the same functionality for C/C++ programs targeted to Xilinx FPGAs.

4 FPGA Architecture

An FPGA is a type of integrated circuit (IC) that can be programmed for different algorithms after fabrication. Modern FPGA devices consist of up to two million logic cells that can be configured to implement a variety of software algorithms. Although the traditional FPGA design flow is more similar to a regular IC than a processor, an FPGA provides significant cost advantages in comparison to an IC development effort and offers the same level of performance in most cases. Another advantage of the FPGA when compared to the IC is its ability to be dynamically reconfigured. This process, which is the same as loading a program in a processor, can affect part or all of the resources available in the FPGA fabric.

The basic structure of an FPGA is composed of the following elements:

- Look-up table (LUT): This element performs logic operations.
- Flip-Flop (FF): This register element stores the result of the LUT.
- Wires: These elements connect elements to one another.
- Input/Output (I/O) pads: These physically available ports get data in and out of the FPGA.

The combination of these elements results in the basic FPGA architecture shown in . Although this structure is sufficient for the implementation of any algorithm, the efficiency of the resulting implementation is limited in terms of computational throughput, required resources, and achievable clock frequency.

Basic FPGA Architecture

Figure 1: Basic FPGA Architecture

Contemporary FPGA architectures incorporate the basic elements along with additional computational and data storage blocks that increase the computational density and efficiency of the device. These additional elements are:

- Embedded memories for distributed data storage
- Phase-locked loops (PLLs) for driving the FPGA fabric at different clock rates
- High-speed serial transceivers
- Off-chip memory controllers
- Multiply-accumulate blocks

The combination of these elements provides the FPGA with the flexibility to implement any software algorithm running on a processor and results in the contemporary FPGA architecture shown in .

Figure 2: Contemporary FPGA Architecture

4.1 Look Up Table (LUT)

The LUT is the basic building block of an FPGA and is capable of implementing any logic function of N Boolean variables. Essentially, this element is a truth table in which different combinations of the inputs implement different functions to yield output values. The limit on the size of the truth table is N , where N represents the number of inputs to the LUT. For the general N -input LUT, the number of memory locations accessed by the table is:

$$2^N$$

which allows the table to implement the following number of functions:

$$2^{2^N}$$

Note: A typical value for N in Xilinx FPGA devices is 6.

The hardware implementation of a LUT can be thought of as a collection of memory cells connected to a set of multiplexers. The inputs to the LUT act as selector bits on the multiplexer to select the result at a given point in time. It is important to keep this representation in mind, because a LUT can be used as both a function compute engine and a data storage element. shows this functional representation of the LUT.

Functional Representation of a LUT as Collection of Memory Cells

Figure 3: Functional Representation of a LUT as Collection of Memory Cells

4.2 Flip-Flop

The flip-flop is the basic storage unit within the FPGA fabric. This element is always paired with a LUT to assist in logic pipelining and data storage. The basic structure of a flip-flop includes a data input, clock input, clock enable, reset, and data output. During normal operation, any value at the data input port is latched and passed to the output on every pulse of the clock. The purpose of the clock enable pin is to allow the flip-flop to hold a specific value for more than one clock pulse. New data inputs are only latched and passed to the data output port when both clock and clock enable are equal to one. shows the structure of a flip-flop.

Structure of a Flip-Flop

Figure 4: Structure of a Flip-Flop

4.3 DSP Block

The most complex computational block available in a Xilinx FPGA is the DSP block, which is shown in . The DSP block is an arithmetic logic unit (ALU) embedded into the fabric of the FPGA, which is composed of a chain of three different blocks. The computational chain in the DSP is composed

of an add/subtract unit connected to a multiplier connected to a final add/subtract/accumulate engine. This chain allows a single DSP unit to implement functions of the form:

$$p = a * (b + d) + c$$

or

$$p + = a * (b + d)$$

Structure of a DSP Block

Figure 5: Structure of a DSP Block

4.4 Storage Elements

The FPGA device includes embedded memory elements that can be used as random-access memory (RAM), read-only memory (ROM), or shift registers. These elements are:

1. Block RAMs (BRAMs),
2. UltraRAM blocks (URAMS),
3. LUTs, and
4. Shift registers (SRLs).

4.4.1 BRAM

The BRAM is a dual-port RAM module instantiated into the FPGA fabric to provide on-chip storage for a relatively large set of data. The two types of BRAM memories available in a device can hold either 18k or 36k bits. The number of these memories available is device specific. The dual-port nature of these memories allows for parallel, same-clock-cycle access to different locations. In terms of how arrays are represented in C/C++ code, BRAMs can implement either a RAM or a ROM. The only difference is when the data is written to the storage element. In a RAM configuration, the data can be read and written at any time during the runtime of the circuit. In contrast, in a ROM configuration, data can only be read during the runtime of the circuit. The data of the ROM is written as part of the FPGA configuration and cannot be modified in any way.

4.4.2 URAM

The UltraRAM blocks are dual-port, synchronous 288 Kb RAM with a fixed configuration of 4,096 bits deep and 72 bits wide. They are available on UltraScale+ Devices and provide 8 times more storage capacity than the BRAM.

4.4.3 LUT

As previously discussed, the LUT is a small memory in which the contents of a truth table are written during device configuration. Due to the flexibility of the LUT structure in Xilinx FPGAs, these blocks can be used as 64-bit memories and are commonly referred to as distributed memories. This is the fastest kind of memory available on the FPGA device, because it can be instantiated in any part of the fabric that improves the performance of the implemented circuit.

4.4.4 Shift Register

The shift register is a chain of registers connected to each other. The purpose of this structure is to provide data reuse along a computational path.

5 FPGA Parallelism Versus Processor Architectures

When compared with processor architectures, the structures that comprise the FPGA fabric enable a high degree of parallelism in application execution. The custom processing architecture generated by the Vivado HLS compiler for a software program presents a different execution paradigm, which must be taken into account when deciding to port an application from a processor to an FPGA.

5.1 Program Execution on a Processor

A processor, regardless of its type, executes a program as a sequence of instructions that translate into useful computations for the software application. This sequence of instructions is generated by processor compiler tools, such as the GNU Compiler Collection (GCC), which transform an algorithm expressed in C/C++ into assembly language constructs that are native to the processor.

Any assembly code can show that even a simple operation, results in multiple assembly instructions. The computational latency of each instruction is not equal across instruction types.

IMPORTANT: The level of effort required by the software engineer in restructuring algorithms to better fit the available processor cache is not required when the same operation is implemented in an FPGA.

5.2 Program Execution on an FPGA

The FPGA is an inherently parallel processing fabric capable of implementing any logical and arithmetic function that can run on a processor. The main difference is that the Vivado HLS compiler, which is used to transform software descriptions into RTL, is not hindered by the restrictions of a cache and a unified memory space. Every code is compiled by Vivado HLS into several LUTs required to achieve the size of the output operand.

NOTE: As a general rule, 1 LUT is equivalent to 1 bit of computation.

LUTs used for a computation of a value are exclusive to that particular operation only. Unlike a processor, where all computations share the same ALU, an FPGA implementation instantiates independent sets of LUTs for each computation in the software algorithm.

In addition to assigning unique LUT resources per computation, the FPGA differs from a processor in both memory architecture and the cost of memory accesses. In an FPGA implementation, the Vivado HLS compiler arranges memories into multiple storage banks as close as possible to the point of use in the operation. This results in an instantaneous memory bandwidth, which far exceeds the capabilities of a processor.

With regard to computational throughput and memory bandwidth, the Vivado HLS compiler exercises the capabilities of the FPGA fabric through the processes of scheduling, pipelining, and dataflow. Although transparent to the user, these processes are integral stages of the software compilation process that extract the best possible circuit-level implementation of the software application.

5.2.1 Scheduling

Scheduling is the process of identifying the data and control dependencies between different operations to determine when each will execute. Vivado HLS analyzes dependencies between adjacent operations as well as across time. This allows the compiler to group operations to execute in the same clock cycle and to set up the hardware to allow the overlap of function calls. The overlap of function call executions removes the processor restriction that requires the current function call to fully complete before the next function call to the same set of operations can begin. This process is called pipelining.

5.2.2 Pipelining

Pipelining is a digital design technique that allows the designer to avoid data dependencies and increase the level of parallelism in an algorithm hardware implementation. The data dependence in the original software implementation is preserved for functional equivalence, but the required circuit is divided into a chain of independent stages. All stages in the chain run in parallel on the same clock cycle. The only difference is the source of data for each stage. Each stage in the computation receives its data values from the result computed by the preceding stage during the previous clock cycle.

5.2.3 Dataflow

Dataflow is another digital design technique, which is similar in concept to pipelining. The goal of dataflow is to express parallelism at a coarse-grain level. In terms of software execution, this transformation applies to parallel execution of functions within a single program.

Vivado HLS extracts this level of parallelism by evaluating the interactions between different functions of a program based on their inputs and outputs. The simplest case of parallelism is when functions work on different data sets and do not communicate with each other. In this case, Vivado HLS allocates FPGA logic resources for each function and then runs the blocks independently. The more complex case, which is typical in software programs, is when one function provides results for another function. This case is referred to as the consumer-producer scenario. Vivado HLS supports two use models for the consumer-producer scenario:

- In the first use model, the producer creates a complete data set before the consumer can start its operation. Parallelism is achieved by instantiating a pair of BRAM memories arranged as memory banks ping and pong. Each function can access only one memory bank, ping or pong, for the duration of a function call. When a new function call begins, the HLS-generated circuit switches the memory connections for both the producer and the consumer. This approach guarantees functional correctness but limits the level of achievable parallelism to across function calls.
- In the second use model, the consumer can start working with partial results from the producer, and the achievable level of parallelism is extended to include execution within a function call. The Vivado HLS-generated modules for both functions are connected through the use of a first in, first out (FIFO) memory circuit. This memory circuit, which acts as a queue in software programming, provides data-level synchronization between the modules. At any point during a function call, both hardware modules are executing their programming. The only exception is that the consumer module waits for some data to be available from the producer before beginning computation. In Vivado HLS terminology, the wait time of the

consumer module is referred to as the interval or initiation interval (II).

6 Basic Concepts of Hardware Design

One of the key differences between a processor and an FPGA is whether the processing architecture is fixed.

With a processor, the computation architecture is fixed, and the job of the compiler is to determine how to best fit the software application in the available processing structures. Performance is a function of how well the application maps to the capabilities of the processor and the number of processor instructions needed for correct execution.

In contrast, an FPGA is similar to a blank slate with a box of building blocks with the processing architecture not fixed. The job of the Vivado HLS compiler is to create a processing architecture from the box of building blocks that best fits the software program. The process of guiding the Vivado HLS compiler to create the best processing architecture requires fundamental knowledge about hardware design concepts. As with processor compilers, the Vivado HLS compiler handles the low-level details of the algorithm implementation into the FPGA logic fabric.

6.1 Clock Frequency

The processor clock frequency is one of the first items to consider when determining the execution platform of a specific algorithm. A commonly used guideline is that a high clock frequency translates into a higher performance execution rate of an algorithm. Although this might be a good first order rule for choosing between processors, it is actually misleading and can lead the designer to make the wrong choice when selecting between a processor and an FPGA.

6.1.1 Instruction execution

The first major difference between a processor and an FPGA is how a software program is executed. A processor is able to execute any program on a common hardware platform. This common platform comprises the core of the processor and defines a fixed architecture onto which all software must be fitted. The compiler, which has a built-in understanding of the processor architecture, compiles the user software into a set of instructions.

Regardless of the type of processor, standard versus specialized, the execution of an instruction is always the same. Each instruction of the user application must go through the following stages:

1. Instruction fetch (IF)
2. Instruction decode (ID)
3. Execute (EXE)
4. Memory operations (MEM)
5. Write back (WB)

The purpose of each stage is summarized below.

Stage	Description
IF	Get the instruction from program memory.

Stage	Description
ID	Decode the instruction to determine the operation and the operators.
EXE	Execute the instruction on the available hardware. In a standard processor, this means the arithmetic logic unit (ALU) or floating point unit (FPU). A specialized processor adds on fixed function accelerators to the capabilities of the standard processor at this stage of instruction processing.
MEM	Fetch data for the next instruction using memory operations.
WB	Write the results of the instruction either to local registers or global memory.

Most modern processors include multiple copies of the instruction execution path and are capable of running instructions with some degree of overlap. Because instructions in a processor usually depend on each other, the overlap between copies of the instruction execution hardware is not perfect. In the best of cases, only the overhead stages introduced by using a processor can be overlapped. The EXE stages, which are responsible for application computation, execute sequentially. The reasons for this sequential execution are related to limited resources in the EXE stage and dependence between instructions.

shows a processor with multiple instructions executing in a semi-parallel order. This is the best case for a processor in which all instructions are executing as quickly as possible. Even in this best case, the processor is limited to only one EXE stage per clock cycle. This means that the user application moves forward by one operation per clock cycle. Even if the compiler determined that all five EXE stages could execute in parallel, the structure of the process would prevent it.

Processor with Multiple Instruction Execution Units

Figure 6: Processor with Multiple Instruction Execution Units

An FPGA does not execute all software on a common computation platform. It executes a single program at a time on a custom circuit for that program. Therefore, changing the user application changes the circuit in the FPGA. Given this flexibility, the Vivado HLS compiler does not need to account for overhead stages in the platform and can find ways of maximizing instruction parallelism. Working with the same assumptions as in , the execution profile of the same software in an FPGA is shown in

FPGA with Multiple Instruction Execution Units

Figure 7: FPGA with Multiple Instruction Execution Units

Based on the comparison of and , the FPGA has a nominal performance advantage of 9x compared to the processor. Actual numbers are always application specific, but FPGAs generally demonstrate at least 10x the performance of a processor for computationally intensive applications.

6.1.2 Power Consumption

Another issue hidden by only focusing on the clock frequency is the power consumption of a software program. The approximation to power consumption is given by:

$$P = (1/2)cFV^2$$

The relationship between power consumption and clock frequency is supported by empirical data, which shows higher power usage in a processor than an FPGA for the same computational workload. By creating a custom circuit per software program, an FPGA is able to run at a lower clock frequency with maximum parallelism between operations and without the instruction interpretation overhead found in a processor.

6.2 Latency and Pipelining

Latency is the number of clock cycles it takes to complete an instruction or set of instructions to generate an application result value. Using the basic processor architecture, the latency of an instruction is five clock cycles. If the application has a total of five instructions, the overall latency for this simple model is 25 clock cycles. That is, the result of the application is not available until 25 clock cycles expire. Application latency is a key performance metric in both FPGAs and processors. In both cases, the problem of latency is resolved through the use of pipelining.

In a processor, pipelining means that the next instruction can be launched into execution before the current instruction is complete. This allows the overlap of overhead stages required in instruction set processing. By overlapping the execution of instructions, the processor achieves a latency of nine clock cycles for the five instruction application.

In an FPGA, the overhead cycles associated with instruction processing are not present. The latency is measured by how many clock cycles it takes to run the EXE stage of the original processor instruction. In the case of FPGA, the latency is one clock cycle. Parallelism also plays an important role in latency. For the full five instruction application, the FPGA latency is one clock cycle, as shown in . With the one clock cycle latency of the FPGA, it might not be clear why pipelining is advantageous. However, the reason for pipelining in an FPGA is the same as in a processor, that is, to improve application performance.

As previously explained, the FPGA is a blank slate with building blocks that must be connected to implement an application. The Vivado HLS compiler can connect the blocks directly or through registers. shows an implementation of the EXE stages without Pipelining.

FPGA Implementation without Pipelining

Figure 8: FPGA Implementation without Pipelining

- **Operation timing** in an FPGA is the length of time it takes a signal to travel from a source register to a sink register.
- For the case of an FPGA circuit, the **clock frequency** is defined as the longest signal travel time between source and sink registers.
- **Pipelining** in an FPGA is the process of inserting more registers to break up large computation blocks into smaller segments. This partitioning of the computation increases the latency in absolute number of clock cycles but increases performance by allowing the custom circuit to run at a higher clock frequency.

6.2.1 Important points

shows the implementation of the processing architecture in after complete pipelining. Complete pipelining means that a register is inserted between each building block in the FPGA circuit.

The addition of registers reduces the timing requirement of the circuit, which results in a maximum clock frequency. In addition, by separating the computation into separate register-bounded regions, each block is allowed to always be busy, which positively impacts the application throughput.

One issue with pipelining is the latency of the circuit. The original circuit of has a latency of one clock cycle at the expense of a low clock frequency. In contrast, the circuit of has a latency of five clock cycles at a higher clock frequency.

FPGA Implementation with Pipelining

Figure 9: FPGA Implementation with Pipelining

IMPORTANT: The latency caused by pipelining is one of the trade-offs to consider during FPGA design.

6.3 Throughput

Throughput is another metric used to determine overall performance of an implementation. It is the number of clock cycles it takes for the processing logic to accept the next input data sample. With this value, it is important to remember that the clock frequency of the circuit changes the meaning of the throughput number.

For example, both and show implementations that require one clock cycle between input data samples. The key difference is that the implementation in requires 10 ns between input samples, whereas the circuit in only requires 2 ns between input data samples.

After the time base is known, it is clear that the second implementation has higher performance, because it can accept a higher input data rate.

Note: The definition of throughput described in this section can also be used when analyzing applications executing on a processor.

6.4 Memory Architecture and Layout

The memory architecture of the selected implementation platform is one of the physical elements that can affect the performance of a software application. Memory architecture determines the upper bound on achievable performance. At some performance point, all applications on either a processor or an FPGA become memory bound regardless of the type and number of available computational resources.

In a processor-based system, the software engineer must fit the application on essentially the same memory architecture regardless of the specific type of processor. This commonality simplifies the process of application migration at the expense of performance. Common memory architecture familiar to software engineers consists of memories that are slow, medium, or fast based on the number of clock cycles it takes to get the data to the processor. These memory classifications are defined below.

Memory Type	Definition
Slow	Mass storage devices such as hard drives
Medium	DDR memories

Memory Type	Definition
Fast	On-chip cache memories of different sizes depending on the specific processor

The memory architecture shown in this table assumes that the user is presented with a single large memory space. Within this memory space, the user allocates and de-allocates regions to store program data. The physical location of data and how it moves between the different levels in the hierarchy is handled by the computation platform and is transparent to the user. In this kind of system, the only way to boost performance is to reuse data in the cache as much as possible.

To achieve this goal, the software engineer must spend large amounts of time looking at cache traces, restructuring the software algorithm to increase data locality, and managing memory allocation to minimize the instantaneous memory footprint of the program.

Although all of these techniques are portable across processors, the results are not. A software program must be tuned for each processor it runs on to maximize performance.

FPGA-based systems can be attached to slow and medium memories but exhibit the greatest degree of differentiation in terms of available fast memories. That is, instead of restructuring the software to best use an existing cache, the Vivado HLS compiler builds a fast memory architecture to best fit the data layout in the algorithm. The resulting FPGA implementation can have one or more internal banks of different sizes that can be accessed independently from one another.

The use of dynamic memory allocation has long been part of the best practice guidelines for processor-based systems due to the underlying fixed memory architecture.

In contrast to this approach, the Vivado HLS compiler builds a memory architecture that is tailored to the application. This tailored memory architecture is shaped both by the size of the memory blocks in the program as well as by how the data is used throughout program execution. Vivado HLS requires that the memory requirements of an application are fully analyzable at compile time.

The benefit of static memory allocation is that Vivado HLS can implement the memory in different ways. Depending on the computation in the algorithm, the Vivado HLS compiler can implement the memory as registers, shift registers, FIFOs, or BRAMs.

6.4.1 Registers

A register implementation of a memory is the fastest possible memory structure. Each independent entity to be stored is embedded into the computation where it is used without the need to address logic or additional delays.

6.4.2 Shift Register

In processor programming terms, a shift register can be thought of as a special case of a queue. The key characteristic of a shift register is that every element can be accessed on every clock cycle. In addition, moving all data items to the next adjacent storage container requires only one clock cycle.

6.4.3 FIFO

A FIFO can be thought of as a queue with a single point of entry and a single point of exit. This kind of structure is typically used to transmit data between program loops or functions. There is no addressing logic involved, and the implementation details are completely handled by the Vivado HLS compiler.

6.4.4 BRAM

A BRAM is a random-access memory that is embedded into the FPGA fabric. A Xilinx FPGA device includes many of these embedded memories. The exact number of memories is device specific. In processor programming terms, this kind of memory can be thought of as a cache with the following limitations:

- Does not implement cache coherency, collision, and cache miss tracking logic typically found in a processor cache.
- Holds its values only as long as the device is powered on (Volatile).
- Supports parallel same cycle access to two different memory locations.

7 Vivado High-Level Synthesis

The Xilinx Vivado High-Level Synthesis (HLS) compiler provides a programming environment similar to those available for application development on both standard and specialized processors. Vivado HLS shares key technology with processor compilers for the interpretation, analysis, and optimization of C/C++ programs. The main difference is in the execution target of the application.

By targeting an FPGA as the execution fabric, Vivado HLS enables a software engineer to optimize code for throughput, power, and latency without the need to address the performance bottleneck of a single memory space and limited computational resources.

Application code targeting the Vivado HLS compiler uses the same categories as any processor compiler. Vivado HLS analyzes all programs in terms of:

- Operations
- Conditional statements
- Loops
- Functions

IMPORTANT: Vivado HLS can compile almost any C/C++ program. The only coding limitation for Vivado HLS is with dynamic language constructs typical in processors with a single memory space. When using Vivado HLS, the main dynamic constructs to consider are memory allocation and pointers.

7.1 Operations

Operations refer to both the arithmetic and logical components of an application that are involved in computing a result value.

- With a processor compiler, the fixed processing architecture means that the user can only affect performance by limiting operation dependency and manipulating memory layout to maximize cache performance.
- In contrast, Vivado HLS is not constrained by a fixed processing platform and builds an algorithm-specific platform based on user input. This allows an HLS designer to affect application performance in terms of throughput, latency, and power.

7.1.0.1 Tradeoff between resources, performance, and application Although an FPGA does not have a fixed processing architecture, each device has a maximum number of building blocks it can sustain. Therefore, the designer can evaluate FPGA resources versus application performance versus the number of applications per device.

7.1.1 Optimizations by vivado HLS

- The scheduling of the operations to reduce the latency is one of the automatic resource optimizations from Vivado HLS.
- For memory operations, Vivado HLS analyzes the banks containing the data and where the value is consumed during computation.
- Another way in which Vivado HLS helps the user control the size of the generated circuit is by providing data types for the sizing of variables. Vivado HLS offers the user access to integer, single precision, and double precision data types.
- Vivado HLS also offers arbitrary precision data types for eliminating inefficiencies caused due to 32-bit and 64-bit data types.
- Arbitrary precision data types reduces the number of resources, the number of levels of logic required to complete an operation, and in the latency of a design.
- Based on the size of operations, Vivado HLS automatically does the optimization by pipelining, or division of computation into smaller register-bound regions.
- Vivado HLS divides large operators into multiple computation stages with a corresponding increase in circuit latency.

7.2 Conditional Statements

Conditional statements are program control flow statements that are typically implemented as if, if-else, or case statements. These coding structures are an integral part of most algorithms and are fully supported by all compilers, including HLS. The only difference between compilers is how these types of statements are implemented.

With a processor compiler, conditional statements are translated into branch operations that might or might not result in a context switch. The introduction of branches disrupts the maximum instruction execution packing by introducing a dependence that affects which instruction is fetched next from memory. This uncertainty results in bubbles in the processor execution pipeline and directly affects program performance.

In an FPGA, a conditional statement does not have the same potential impact on performance as in a processor. Vivado HLS creates all the circuits described by each branch of the conditional state-

ment. Therefore, the runtime execution of a conditional software statement involves the selection between two possible results rather than a context switch.

7.3 Loops

Loops are a common programming construct for expressing iterative computation. HLS fully supports loops. The default behavior of HLS is to execute loops in the same schedule as a processor but, HLS can parallelize or pipeline the iterations of a loop to reduce computation latency and increase the input data rate. HLS creates the hardware for the algorithm, it can alter the execution profile of a loop by following optimizations:

7.3.1 Optimizations by vivado HLS

1. To reduce iteration latency, Vivado HLS uses **Operator Parallelization** to the loop iteration body.
2. **Loop Iteration Pipelining** requires user input, as it affects the resource consumption and input data rates of the FPGA implementation.

The user controls the level of iteration pipelining by setting the **Loop Initialization Interval (II)**. The II of a loop specifies the number of clock cycles between the start times of consecutive loop iterations.

To achieve iteration pipelining, HLS analyzes the data dependencies and resource contentions between loop iterations 0 and 1 and automatically resolves issues as follows:

- To resolve data dependencies, HLS alters one of the operations in the loop body or queries the user for algorithm changes.
- To resolve resource contentions, HLS instantiates more copies of the resource or queries the user for algorithm changes.

7.4 Functions

Functions are a programming hierarchy that can contain operators, loops, and other functions. The treatment of functions in both HLS and processor compilers is similar to that of loops.

In HLS, the main difference between loops and functions is related to terminology. HLS can parallelize the execution of both loops and functions. To avoid potential confusion when working with HLS, the parallelization of function call execution is referred to as **Dataflow Optimization**.

The dataflow optimization instructs HLS to create independent hardware modules for all functions at a given level of program hierarchy. These independent hardware modules are capable of concurrent execution and self-synchronize during data transfer.

7.5 Dynamic Memory Allocation

Dynamic memory allocation is one of the memory management techniques available in the C and C++ programming languages. In this method, the user can allocate as much memory as necessary during program runtime. The size of the allocated memory can vary between executions of the program and is allocated from a central physical pool of memory. The function calls typically associated with dynamic memory allocation are shown below:

C	C++
malloc()	new()
calloc()	delete()
free()	

As FPGA does not have a fixed memory architecture, HLS synthesizes the memory architecture based on the unique requirements of the algorithm. Therefore, all code provided to the HLS compiler for implementation in an FPGA must use compile time analyzable memory allocation only.

To aid the user in ensuring that all code provided to HLS is synthesizable, the compiler executes a coding compliance pass before analyzing the design. This code compliance pass flags all coding styles that are not suitable for HLS. It is the responsibility of the user to manually change the code and remove all instances of dynamic memory allocation.

Note: HLS cannot synthesize code that includes any of the dynamic memory allocation keywords even if the allocation is constant.

```
// Dynamic Memory Allocation in C/C++
int *A = malloc(10*sizeof(int));
// This code is not synthesized in HLS.
```

There are two possible methods of modifying this code to comply with HLS.

7.5.0.1 HLS-Compliant Automatic Memory Allocation HLS implements this memory style in strict accordance with the behavior stipulated by C/C++. This means that the memory created to store array A only stores valid data values during the duration of the function call containing this array. Therefore, the function call is responsible for populating A with valid data before each use.

7.5.0.2 HLS-Compliant Static Memory Allocation The behavior for this type of memory allocation dictates that the contents of array A are valid across function calls until the program is completely shut down. When working with HLS, the memory that is implemented for array A contains valid data as long as there is power to the circuit.

```
// HLS-Compliant Automatic Memory Allocation
int A[10];

// HLS-Compliant Static Memory Allocation
static int A[10];
```

Both automatic and static memory allocation techniques can increase the overall software memory footprint of an algorithm running on a processor. When specifying algorithms in C/C++ for FPGA implementation, the most important consideration is the overall goal of the user application. That is, the main goal when compiling to an FPGA is not creating the best software algorithm implementation. Instead, when using tools like HLS, the goal is to capture the algorithm in a way that allows the tool to infer the best possible hardware architecture, which results in the best possible implementation.

7.6 Pointers

A pointer is an address to a location in memory. The HLS compiler supports pointer usage that can be completely analyzed at compile time. An analyzable pointer usage is usage that can be fully expressed and computed in a pen and paper computation without the need for runtime information.

```
// Valid coding style in which pointers are used to access a
memory.
int A[10];
int *pA;
pA = A;
```

Another supported model for memories and pointers is in accessing external memory. When using HLS, any pointer access on function parameters implies either a variable or an external memory. HLS defines an external memory as any memory outside of the scope of the compiler-generated RTL. This means that the memory might be located in another function in the FPGA or in part of an off-chip memory, such as DDR.

8 Software Verification and Vivado HLS

8.1 Software Test Bench

Verification of any HLS-generated module requires a software test bench. The software test bench serves the following important functions:

- To prove that the software targeted for FPGA implementation runs and does not create a segmentation fault.
- To prove the functional correctness of the algorithm.

8.1.0.1 Segmentation faults Segmentation faults are an issue in HLS as they are in any other compiler. However, there is a difference in how the coding error that caused the issue is detected.

In a processor-based execution, segmentation faults are caused by a program trying to access a memory location that is not known to the processor. Detection of this error is relatively straightforward at runtime based on the following sequence of events:

1. Processor detects a memory access violation and notifies the operating system (OS).
2. OS signals the program or process causing the error.
3. After receiving the error signal from the OS, the program terminates and generates a core dump file for analysis.

In an HLS-generated implementation, it is difficult to detect a segmentation fault, because there is no processor and no operating system monitoring program execution. The only indicator of a segmentation fault is the appearance of incorrect result values generated by the circuit.

RECOMMENDED: When working with HLS, it is recommended that the designer ensure that the software test bench compiles and executes the function without issues on a processor. This guarantees that the HLS-generated implementation will not result in a segmentation fault.

The other purpose of the software test bench is to prove the functional correctness of an algorithm targeted towards FPGA execution. For the generated hardware implementation, the HLS compiler guarantees only functional equivalence with the original C/C++ code. Therefore, the existence of a good software test bench is required to minimize efforts in hardware verification and validation.

A good software test bench is characterized by the execution of thousands or millions of data set tests on the software implementation of an algorithm. This allows the designer to assert with a high level of confidence that the algorithm was captured properly. However, even with many test vectors, it is sometimes still possible to detect errors in the HLS-generated output during hardware verification of an FPGA design. Detecting functional errors during hardware verification means that the software test bench was incomplete. Applying the offending test vector to the C/C++ execution reveals the incorrect statement in the algorithm.

IMPORTANT: Errors must not be fixed directly in the generated RTL. Any issues with functional correctness are a direct result of the functional correctness of the software algorithm.

TIP: The software test bench used to exercise an algorithm targeted for FPGA implementation with HLS does not have any coding style restrictions. The software engineer is free to use any valid C/C++ coding style or construct to thoroughly test the functional correctness of an algorithm.

8.2 Code Coverage

Code coverage indicates what percentage of the statements in a design are exercised by the test bench code. This metric, which can be generated by tools like gcov, gives an idea of the quality of the test vectors used to exercise the algorithm.

At a minimum, a test bench must receive a 90% code coverage score to be considered an adequate test of an algorithm. This means that the test vectors trigger all branches in case statements, conditional if-else statements, and for loops. Aside from the overall coverage metric, the report generated by code coverage tools provide insight into which parts of a function are executed and which are not.

Running gcov requires that the code is compiled with additional flags that generate the information needed for profiling the execution of a program. Example gcov command on example code named example.c

```
gcc -fprofile-arcs -ftest-coverage example.c
./a.out
gcov example.c
```

8.3 Uninitialized Variables

Uninitialized variables are a result of a poor coding style in which the designer does not initialize variables to 0 at the point of declaration. For example,

```
int A;
int B;
...
A = B * 100;
```

Although some processor compilers resolve the problem by automatically assigning 0 at the point of declaration, HLS does not use this type of solution. HLS assumes that any undefined behavior in

the user code can be optimized out of the resulting implementation. This triggers an optimization cascade effect that can reduce the circuit to nothing. A user can detect this type of error by noticing the empty RTL files for the generated implementation.

A better way to detect this type of error is to use code analysis tools, such as **valgrind** & **Coverity**. Both of these tools flag uninitialized variables in the user program. Like all software quality issues, uninitialized variables must be resolved before the code is compiled with HLS.

8.4 Out-of-Bounds Memory Access

In HLS, memory accesses are expressed either as operations on an array or as operations on an external memory through a pointer. In the case of out-of-bounds memory access, the focus is on arrays that are converted into memory blocks by HLS.

In a processor compiler, this type of address overflow triggers the address counter to reset to 0. Although the result is functionally incorrect, this kind of error does not usually result in a program crash.

With HLS, accessing an invalid address triggers a series of events that result in an irrecoverable runtime error in the generated circuit. Because the HLS implementation assumes that the software algorithm was properly verified, error recovery logic is not included in the generated FPGA implementation.

Therefore, an invalid memory address is generated by the implementation of the code to the BRAM resource element. The BRAM then issues an error condition that is not expected by the HLS implementation, and the error is left unattended. The unattended error from the BRAM causes the system to hang and can only be resolved with a device reboot.

To catch cases like this before circuit compilation, it is recommended that the tool is executed through a dynamic code checker such as valgrind. Valgrind is a suite of tools designed to check and profile the quality of a C/C++ program. The valgrind Memcheck tool executes a compiled C/C++ program and monitors all memory operations during execution. This tool flags the following critical issues:

1. Use of uninitialized variables
2. Invalid memory access requests

RECOMMENDED: Before using HLS to compile a software function for FPGA execution, it is recommended that all of the issues flagged by a dynamic code checker are resolved by the designer.

8.5 Co-Simulation

Tools for C/C++ program analysis and functionality testing catch most of the issues that affect an HLS implementation. However, these tools are unable to verify whether a sequential C/C++ program maintains functional correctness after parallelization. This issue is resolved in the HLS compiler by the process of co-simulation.

Co-simulation is a process in which the generated FPGA implementation is exercised by the same C/C++ test bench used during software simulation. HLS handles the communication between the C/C++ test bench and the generated RTL in a manner that is transparent to the user. As part of this process, HLS invokes a hardware simulator, such as the Vivado simulator, to emulate how the RTL will function on the device. The main purpose of this simulation is to check that

the parallelization guidance provided by the user did not break the functional correctness of the algorithm.

By default, HLS obeys all algorithm dependencies before parallelization to ensure functional equivalence with the original C/C++ representation. In cases where an algorithm dependence cannot be fully analyzed, HLS takes a conservative approach and obeys dependence. This can lead the compiler to generate a conservative implementation that does not achieve the target performance goals of the application.

When using co-simulation, it is important to remember that this is a simulation of parallel hardware being executed on a processor. Therefore, it is approximately 10,000 times slower than C/C++ simulation.

It is also important to remember that the purpose of co-simulation is not to verify the functional correctness of an algorithm. Instead, the purpose is to check that the algorithm was not broken by user guidance to the HLS compiler.

8.6 When C/C++ Verification Is Not Possible

There are still some cases where the C/C++ representation of an algorithm cannot be fully verified before HLS compilation.

For example, the usage of the volatile keyword, which is common in device driver development, alerts the compiler that the pointers are connected to storage elements that might change during the execution of the function. This kind of pointer must be read or written every time it is specified in the source code. Traditional compiler optimizations to coalesce pointer accesses are also turned off by the volatile keyword.

The issue with volatile data is that the behavior of the code cannot be fully verified in a C/C++ simulation. C/C++ simulation does not have the ability to change the value of a pointer in the middle of the execution of the function under test. Therefore, this type of code can only be fully verified in an RTL simulation after HLS compilation. The user must write an RTL test bench to test the generated circuit in all possible cases for each volatile pointer in the C/C++ source. The use of co-simulation is not applicable in this case, because it is limited by the test vectors that can be used in a C/C++ simulation.

9 Design Example: Application Running on a Zynq-7000 SoC

This design example shows how to take processor code and transform it into an application that runs on a Zynq-7000 SoC. This example walks through the following steps in the migration process:

- Analyzing and partitioning the processor code
- Compiling the program in Vivado HLS
- Composing the system in Vivado IP integrator
- Connecting processor code and FPGA fabric functions

9.1 Analyzing and partitioning the processor code

Most software applications targeted for a Zynq-7000 device begin as applications executing on either a standard x86 processor or a DSP processor. Therefore, the first step in migrating a design

is to compile the program for the Arm Cortex-A9 processor and analyze its performance. The performance analysis data of a program running on the Arm processor guides the designer in choosing how to partition the original code between processor and FPGA fabric.

Post compilation to the Arm processor, there are two ways of analyzing program performance:

- **Measuring timing:** This method involves instrumenting the code with timers and timing the execution of each sub-function on the processor.
- **Using code profiling tools:** This less intrusive method uses tools, such as gprof, to measure the amount of time spent on a function and to provide statistics on the number of times the function is called.

Based on the results of gprof the decision is made where the functions are implemented. After a function is marked for FPGA implementation, the function ports must be analyzed to determine the most suitable hardware interface.

9.2 Compiling the program in Vivado HLS

After identifying the functions to run in the FPGA fabric, the designer prepares the source code for Vivado HLS compilation.

RECOMMENDED: When working with multiple projects or modules, it is recommended that the source code is separated into different files. This simple technique prevents issues with one module compilation affecting the other module in the design.

HLS compilation can be controlled using a Tool Command Language (Tcl) script file. A Tcl script file, which is analogous to a compilation Makefile, instructs the compiler which function to implement and FPGA device to target. The script is divided into the following sections:

Project setup This section includes the source files and the name of the function to be compiled. Guiding the Vivado HLS compiler is an iterative process of applying directives or pragmas to the design source code. Each successive refinement of a design is called a solution. All projects have at least one solution.

Solution setup This section establishes the clock frequency and device for which the software function is compiled. If the designer is guiding the compiler through the use of directives, the solution directives are included in this section of the script.

Compilation setup This section drives the RTL generation and packaging. The assembly of HLS programs into an FPGA device application requires the use of the Vivado IP integrator, which is a system composition tool. IP integrator requires modules to be packed in the equivalent of a software object file.

Interface pragmas define how the module is connected in the FPGA fabric. The definition process is separated into interface behavior and interface mapping.

9.3 Composing the system in Vivado IP integrator

Vivado IP integrator is a Xilinx FPGA design tool for system composition. One use of this tool is to take the blocks generated by the HLS compiler and connect them into the processing platform that executes the user application. In software development terms, IP integrator is analogous to

a linker that combines all program objects into a single bitstream. A bitstream is the binary file used to program the FPGA fabric.

9.4 Connecting processor code and FPGA fabric functions

After the FPGA fabric programming binary is created in IP integrator, the designer must create the software that runs on the processor. The purpose of this software is to initialize the FPGA fabric functions, launch execution, and receive results from the fabric. For the overall application to be functionally equivalent to the original processor code, each function running in the FPGA fabric requires the following functionality in the code running on the Arm Cortex-A9 processor:

- Address mapping
- Initialization
- Start function
- Interrupt service routine (ISR)
- Interrupt registration in the processor exception table
- New main function to run the system

10 Verification of a Complete Application

In FPGA design, a complete application refers to a hardware system that implements the functionality captured by the software representation of a design. There are two main categories of systems that can be built on an FPGA using the Vivado HLS compiler:

1. Standalone Compute Systems
2. Processor-Based Systems

10.1 Standalone compute systems

The standalone compute system is an FPGA implementation created by one or more HLS-generated modules connected together to implement a software application. In these types of systems, the configuration of the algorithm is fixed and loaded during device configuration. The modules generated by the HLS compiler are connected to external FPGA pins for data transmit and receive transactions. This is the easiest kind of system to verify. The verification of a standalone system is divided into the following stages:

1. Module verification
2. Connectivity verification
3. Application verification
4. Device validation

10.1.1 Module Verification

Module verification of an HLS-generated block is covered in section "Software Verification and Vivado HLS". After the block is fully verified for functional correctness in both software and

co-simulation, the designer must test the block for in system error tolerance.

Both software simulation and co-simulation are focused on testing the functional correctness of an algorithm in isolation. That is, the algorithm and compiled module are tested to ensure correct functionality when all inputs and outputs are handled in an ideal manner. This thorough level of testing helps to ensure correctness after data is supplied to the module. It also helps to reduce the verification burden of later stages by eliminating the internal processing core of a module as a possible source of error. **The only module-level issue that is not handled by this methodology is verification that the module can recover fully from incorrect handshaking at its interfaces.**

In-system testing tests how the HLS-generated module reacts to incorrect toggling of its input and output ports. The purpose of this testing is to eliminate I/O behavior as an error source that can crash or otherwise adversely affect the module under test. The types of improper use cases tested in this methodology are:

- Erratic clock signal toggling
- Reset operation and random reset pulsing
- Input ports receiving data at different rates
- Output ports being sampled at different rates
- Interface protocol violations

These tests, which are examples of system-level behavior, ensure that the HLS-generated module functions as expected under all circumstances. The amount of testing required at this stage depends on the types of interfaces and the integration methodology. By using HLS default settings to generate AXI-compliant interfaces, the designer can avoid writing an exhaustive test bench of incorrect system-level behavior. AXI-compliant interfaces are fully tested and verified by the developers of the HLS compiler.

10.1.2 Connectivity Verification

Connectivity verification is a sequence of tests to check that the modules in an application are properly connected to each other. As with module verification, the amount of testing required depends on the system integration methodology. FPGA design tool assistance is provided in both the Xilinx System Generator and Vivado IP integrator design flows. These graphical module connection tools handle all the aspects related to module connection.

The manual integration flow requires the user to write an application top-level module in RTL and manually connect the RTL ports of every module that makes up an application. This is the most error-prone flow and must be verified. The amount of testing required can be decreased by using HLS compiler defaults and generating AXI interfaces for every module port.

For systems built around AXI interfaces, the connectivity can be verified through the use of a bus functional model (BFM). The BFM provides the Xilinx-verified behavior of AXI buses and point-to-point connections. These models can be used for traffic generators, which help prove the correct connection of HLS-generated modules as part of an RTL simulation.

IMPORTANT: It is important to remember that the purpose of this simulation is only to check connectivity and the proper flow of data through the system. The connectivity verification step

does not verify the functional correctness of the application.

10.1.3 Application Verification

Application verification is the final step before running the application on the FPGA device. Application verification focuses on checking that the original software model matches the results of the FPGA implementation. If the application is composed of a single HLS-generated module, this stage is the same as module verification. In cases where the application is composed of two or more HLS-generated modules, the verification process starts with the original software model. The designer must extract application input and output test vectors from the software model to be used in an RTL simulation. Because the construction of the hardware implementation is verified in multiple stages, the application verification does not need to be an exhaustive simulation. The simulation can run as many test vectors as needed for the designer to feel confident in the FPGA implementation.

10.1.4 Device Validation

After an application is assembled in RTL using either automated or manual integration flows, the design goes through an additional compilation stage to generate the binary or bitstream required to program the FPGA. In the terminology of FPGA design, the compilation of RTL into a bitstream is referred to as logic synthesis, implementation, and bitstream generation. Once the bitstream is generated, the FPGA device can be programmed. The application is validated after the hardware runs correctly for an amount of time specified by the designer.

10.2 Processor-based systems

For the module and connectivity verification stages, the verification flow for a processor-based system is the same as the standalone system. The major difference is that a portion of the application is running on the processor.

This partitioning presents a verification challenge that can be addressed through the use of the following technologies:

- Hardware in the loop (HIL) verification
- Virtual platform (VP) verification

10.2.1 Hardware in the Loop (HIL) Verification

HIL verification is a verification methodology in which the simulation of part of the system under test is executed in the FPGA fabric. The main advantages of HIL verification versus verification are:

- No simulation inconsistencies between a processor model and the actual processor
- Code running on the processor is executed at the speed of the FPGA device
- Full visibility into how each generated module operates through RTL simulation

When using HIL verification, it is important to remember the performance characteristics of this technology. Although the processor code runs in the actual hardware, the FPGA fabric is fully simulated on the designer's workstation. HIL verification is only recommended for verifying the

major interactions between the processor and the FPGA fabric, not every use case in the application. The key application behaviors to check with HIL verification are:

- Vivado HLS driver integration into the processor code
- Writing configuration parameters from the PS to the PL
- Interrupt from the PL to the PS

10.2.2 Virtual Platform(VP) Verification

Virtual platform technology is an established method of overlapping software and hardware development. A virtual platform is a software simulation of both the application and the hardware platform on which it runs. The models used for the PL portion of the design can be in C, C++, SystemC, or RTL. This simulation platform can be used as a proxy for the other recommended verification stages with varying degrees of fidelity to the hardware implementation.

In the fastest use case of the virtual platform, the application modules targeted to the PL are simulated from the C/C++ source code provided to the Vivado HLS compiler. This setup results in a functionally correct simulation that allows the designer to test the algorithm for correct computation. As modules are optimized and compiled with Vivado HLS, the generated RTL can replace the software version of the module to enable connectivity testing and timing driver simulation.

IMPORTANT: It is important to remember that adding RTL modules impacts the runtime on the virtual platform and slows down execution.

10.2.3 Device Validation

The purpose of device validation is to check all the application use cases in which the processor interacts with the FPGA fabric. As in the case of standalone execution, this process involves running the complete application for a certain amount of time on the FPGA/SoC. The purpose of this test is to check all the application corner cases with regard to the interaction between the PS and PL portions of the design.

11 Design Principles for Software Programmers

11.1 Introduction

The discussion in this section is tool-agnostic and the concepts introduced are common to most HLS tools.

- **Throughput** is defined as the number of specific actions executed per unit of time or results produced per unit of time.
- **Performance** is defined as higher throughput with low power consumption. Lower power consumption is just as important as higher throughput.

11.2 Computer Architecture

The von Neumann architecture is the basis of almost all computing done today even though it was designed more than 7 decades ago. This architecture was deemed optimal for a large class of applications and has tended to be very flexible and programmable. However, as application

demands started to stress the system, CPUs began supporting the execution of multiple processes. Multithreading and/or Multiprocessing can include multiple system processes (executing two or more programs at the same time), or it can consist of one process that has multiple threads within it. Multi-threaded programming using a shared memory system became very popular as it allowed the software developer to design applications by keeping parallelism in mind but with a fixed CPU architecture.

But when multi-threading and the ever-increasing CPU speeds could no longer handle the data processing rates, multiple CPU cores and hyperthreading were used to improve throughput as shown in the figure on the right.

This general purpose flexibility comes at a cost in terms of power and peak throughput.

To achieve higher throughput, the workload must be closer to memory, and/or into specialized functional units. So the new challenge is to design a new programmable architecture in such a way that you can maintain just enough programmability while achieving higher performance and lower power costs.

A field-programmable gate array (FPGA) provides for this kind of programmability and offers enough memory bandwidth to make this a high-performance and lower power cost solution. Unlike a CPU that executes a program, an FPGA can be configured into a custom hardware circuit that will respond to inputs in the same way that a dedicated piece of hardware would behave.

Reconfigurable devices such as FPGAs contain computing elements of extremely flexible granularities, ranging from elementary logic gates to complete arithmetic-logic units such as DSP blocks. At higher granularities, user-specified composable units of logic called kernels can then be strategically placed on the FPGA device to perform various roles. This characteristic of reconfigurable FPGA devices allows the creation of custom macro-architectures and gives FPGAs a big advantage over traditional CPUs/GPUs in utilizing application-specific parallelism. Computation can be spatially mapped to the device, enabling much higher operational throughput than processor-centric platforms. Today's latest FPGA devices can also contain processor cores (Arm-based) and other hardened IP blocks that can be used without having to program them into the programmable fabric.

11.3 Three Paradigms for Programming FPGAs

While FPGAs can be programmed using lower-level Hardware Description Languages (HDLs) such as Verilog or VHDL, there are now several High-Level Synthesis (HLS) tools that can take an algorithmic description written in a higher-level language like C/C++ and convert it into lower-level hardware description languages such as Verilog or VHDL. This can then be processed by downstream tools to program the FPGA device. The main benefit of this type of flow is that you can retain the advantages of the programming language like C/C++ to write efficient code that can then be translated into hardware.

A program written in C/C++ is essentially written for the von Neumann style of architecture where each instruction is executed sequentially. In order to achieve high performance, the HLS tool must infer parallelism in the sequential code and exploit it to achieve greater performance.

Many coding techniques (such as RTTI, recursion, and dynamic memory allocation) have no direct equivalency in hardware. Hence, arbitrary, off-the-shelf software cannot be efficiently converted into hardware. At a bare minimum, such software needs to be examined for non-synthesizable

constructs and the code needs to be refactored to make it synthesizable. Even the automatically converted (or synthesized) program with acceptable quality of results (QoR), will require additional work to help the HLS tool achieve the desired performance goals.

11.3.1 Producer-Consumer Paradigm

In a multithreaded program, there is usually a master thread that performs some initialization steps and then forks off a number of child threads to do some parallel computation and when all the parallel computation is done, the main thread collates the results and writes to the output.

Similarly, in FPGAs, same fork/join type of parallelism is applied. A key pattern for throughput on FPGAs is the producer-consumer paradigm. The producer-consumer paradigm is applied to a sequential program and it is converted to extract functionality that can be executed in parallel to improve performance. The goal is to decompose the program in such a way that you can identify tasks that can potentially execute in parallel and therefore increase the throughput of the system.

The first step is to understand the program workflow and identify the independent tasks or functions. Two independent tasks need not wait to happen sequentially instead, they can start the execution with interleaving/overlapping. This is referred as Producer-Consumer Paradigm. On FPGAs, due to the custom architecture, the producer and consumer threads can be executed simultaneously with little or no overhead leading to a considerable improvement in throughput unlike a regular CPU.

11.3.2 Streaming Data Paradigm "Dataflow"

A stream is an important abstraction: it represents an unbounded, continuously updating data set, where unbounded means "of unknown or of unlimited size". A stream can be a sequence of data (scalars or buffers) flowing unidirectionally between a source (producer) process and a destination (consumer) process.

Decomposing your algorithm into producer-consumer relationships that communicate by streaming data through the network has several advantages.

- It lets the programmer define the algorithm in a sequential manner and the parallelism is extracted through other means (such as by the compiler).
- Complexities like synchronization between the tasks etc. are abstracted away.
- It allows the producer and the consumer tasks to process data simultaneously, which is key for achieving higher throughput.
- Another benefit is cleaner and simpler code.

11.3.3 Pipelining Paradigm

Another optimization that allows for finer-grained parallelism is pipelining. Pipelining is a process which enables parallel execution of program instructions.

Pipelining does not decrease the latency, that is, the total time for one item to go through the whole system. It does however increase the system's throughput, that is, the rate at which new items are processed after the first one.

- Delay associated with the number of clock cycles lost before the first valid output is delivered is referred to as *Iteration Latency*.
- *Initiation Interval (II)* of the pipeline is defined as the time taken by the design between delivering two consecutive outputs post Iteration Latency or first output.
- The overall time taken to deliver all the outputs through iterations is referred to as the *Total Latency* of the pipeline.

$$\text{Total Latency} = \text{Iteration Latency} + \text{II} * (\text{iterations} - 1)$$

Pipelining is a classical micro-level architectural optimization that can be applied to multiple levels of abstraction. Similar to task-level pipelining with the producer-consumer paradigm, pipelining applies to the instruction-level as well. This is in fact key to keeping the producer-consumer pipelines (and streams) filled and busy. The producer-consumer pipeline will only be efficient if each task produces/consumes data at a high rate, and hence the need for the instruction-level pipelining (ILP).

11.4 Combining the Three Paradigms

Functions and loops are the main focus of most optimizations in the user's program. Each function can be converted into a specific hardware component. Each such hardware component can be instantiated in the hardware design. Each hardware component will in turn be composed of many smaller predefined components that typically implement basic functions such as add, sub, and multiply etc. Functions may call other functions although recursion is not supported.

Functions that are small and called less often can be also inlined into their callers just like how software functions can be inlined. In this case, the resources needed to implement the function are subsumed into the caller function's component which can potentially allow for better sharing of common resources. Constructing your design as a set of communicating functions lends to inferring parallelism when executing these functions.

An inline function is a function for which the compiler copies the code from the function definition directly into the code of the calling function rather than creating a separate set of instructions in memory. This eliminates call-linkage overhead and can expose significant optimization opportunities.

Loops are one of the most important constructs in your program. Since the body of a loop is iterated over a number of times, this property can be easily exploited to achieve better parallelism. There are several transformations (such as *pipelining* and *unrolling*) that can be made to loops and loop nests in order to achieve efficient parallel execution. These transformations enable both memory-system optimizations as well as mapping to multi-core and SIMD execution resources.

In multiple cases, programs are expressed as operations over large data structures with data dependencies across the iterations of the loop. Such data dependencies impact the parallelism achievable in the loop. In many such cases, the code must be restructured such that loop iterations can be executed efficiently and in parallel on modern parallel platforms.

An efficient technique to improve throughput and reuse computational resources is to pipeline operators, loops, and/or functions.

The three paradigms show how parallelism can be achieved in the design without needing the complexities of multi-threading and/or parallel programming languages.

11.5 Conclusion - A Prescription for performance

Checklist of high-level actions for achieving performance on reconfigurable FPGA platforms:

- Software written for CPUs and software written for FPGAs is fundamentally different.
- Right from the start of your project, establish a flow that can functionally verify the source code changes that are being made. Testing the software against a reference model or using golden vectors are common practices.
- **(Producer - Consumer Paradigm:)** Focus first on the macro-architecture of your design.
- Visualize parallelism with function execution timeline and compare with the final achieved results.
- Only start coding or refactoring your program once you have the macro-architecture and the activity timeline established.
- As a general rule, the HLS compiler will only infer task-level parallelism from function calls.
- **Data flow:** Decompose/partition the original algorithm into smaller components that talk to each other via streams (Visualize the data flow).
 - Smaller modular components have the advantage that they can be replicated when needed to improve parallelism.
 - Avoid having communication channels with very wide bit-widths. Decomposing such wide channels into several smaller ones will help implementation on FPGA devices.
 - Large functions (written by hand or generated by inlining smaller functions) can have nontrivial control paths that can be hard for tools to process.
 - Smaller functions with simpler control paths will aid implementation on FPGA devices.
 - Aim to have a single loop nest (with either fixed loop bounds that can be inferred by HLS tool, or by providing loop trip count information by hand to the HLS tool) within each function. This greatly facilitates the measurement and optimization of throughput. (Not be applicable for all designs)
 -
- **Throughput:** Have an overall vision about what rates of processing will be required during each phase of your design is important. Knowing this will influence how you write your application for FPGAs.
 - Try to locate the critical path(s) in the design a potential bottleneck.
 - HLS GUI tools and/or the simulation waveform viewer can be used to visualize such throughput issues.
 - Stream-based communication allows consumers to start processing as soon as producers start producing which allows for overlapped execution (which in turn increases parallelism and throughput).
 - In order to keep the producer and consumer tasks running constantly without any hiccups, optimize the execution of each task to run as fast as possible using techniques such as pipelining and the appropriate sizing of streams.

- Think about the granularity (and overhead) of the streaming channels with respect to synchronization.
- PIPO and FIFO : The usage of PIPO channels allows you to overlap task execution without the fear of deadlock while explicit manual streaming FIFO channels allow you to start the overlapped execution sooner (than PIPOs) but require careful adjustment of FIFO sizes to avoid deadlocks.
- Locate and remove un-synthesizable C/C++ coding styles.
- Use the reports generated by the HLS compiler to guide the optimization process.
- Another important aspect to consider is the interface of accelerated function or kernel. The interface of the kernel to the outside world is an important element of the eventual system design.

12 Introduction to Vitis/Vivado HLS

Vitis/Vivado HLS is a high-level synthesis tool that allows C, C++, and OpenCL functions to become hardwired onto the device logic fabric and RAM/DSP blocks. Vitis/Vivado HLS implements hardware kernels in the Vitis application acceleration development flow and uses C/C++ code for developing RTL IP for Xilinx device designs in the Vivado Design Suite.

In the Vitis application acceleration flow, the Vitis/Vivado HLS tool automates much of the code modifications required to implement and optimize the C/C++ code in programmable logic and to achieve low latency and high throughput. The inference of required pragmas to produce the right interface for your function arguments and to pipeline loops and functions within your code is the foundation of Vitis/Vivado HLS in the application acceleration flow. Vitis/Vivado HLS also supports customization of your code to implement different interface standards or specific optimizations to achieve your design objectives.

Following is the Vitis/Vivado HLS design flow:

1. Compile, simulate, and debug the C/C++ algorithm.
2. View reports to analyze and optimize the design.
3. Synthesize the C algorithm into an RTL design.
4. Verify the RTL implementation using RTL co-simulation.
5. Package the RTL implementation into a compiled object file (.xo) extension, or export to an RTL IP.

12.1 High-Level Synthesis Benefits

High-level synthesis bridges hardware and software domains, providing the following benefits:

- Improved productivity for hardware designers:
Hardware designers can work at a higher level of abstraction while creating high-performance hardware.

- Improved system performance for software designers:
Software developers can accelerate the computationally intensive parts of their algorithms on a new compilation target, the FPGA.

Using a high-level synthesis design methodology allows you to:

Develop algorithms at the C-level Work at a level that is abstract from the implementation details, which consume development time.

Verify at the C-level Validate the functional correctness of the design more quickly than with traditional hardware description languages.

Control the C synthesis process through optimization directives Create specific high-performance hardware implementations.

Create multiple implementations from the C source code using optimization directives Explore the design space, which increases the likelihood of finding an optimal implementation.

Create readable and portable C source code Retarget the C source into different devices as well as incorporate the C source into new projects.

12.2 High-Level Synthesis Basics

High-level synthesis includes the following phases:

- Scheduling Determines which operations occur during each clock cycle based on:
 - Length of the clock cycle or clock frequency.
 - Time it takes for the operation to complete, as defined by the target device.
 - User-specified optimization directives.
 - Operation’s dependencies.
 - Available resource allocation.

If the clock period is longer or a faster FPGA is targeted, more operations are completed within a single clock cycle, and all operations might complete in one clock cycle. Conversely, if the clock period is shorter or a slower FPGA is targeted, high-level synthesis automatically schedules the operations over more clock cycles, and some operations might need to be implemented as multicycle resources.

- Binding Determines which hardware resource implements each scheduled operation. To implement the optimal solution, high-level synthesis uses information about the target device.
- Control logic extraction Extracts the control logic to create a finite state machine (FSM) that sequences the operations in the RTL design.

13 Vitis/Vivado HLS Process Overview

Vitis/Vivado HLS is project based and can contain multiple variations called "solutions" to drive synthesis and simulation. Each solution can target either the Vivado IP flow, or the Vitis Kernel flow. Based on the target flow, each solution will specify different constraints and optimization directives.

The following are the synthesis, analysis, and optimization steps in the typical design flow:

1. Create a new Vitis/Vivado HLS project.
2. Verify the source code with C simulation.
3. Run high-level synthesis to generate RTL files.
4. Analyze the results by examining latency, initiation interval (II), throughput, and resource utilization.
5. Optimize and repeat as needed.
6. Verify the results using C/RTL Co-simulation.

Vitis/Vivado HLS implements the solution based on the target flow, default tool configuration, design constraints, and any optimization pragmas or directives you specify. You can use optimization directives to modify and control the implementation of the internal logic and I/O ports, overriding the default behaviors of the tool.

The C/C++ code is synthesized as follows:

- Top-level function arguments synthesize into RTL I/O port interfaces automatically by Vitis/Vivado HLS.
- Sub-functions of the top-level C/C++ function synthesize into blocks in the hierarchy of the RTL design.
 - The final RTL design includes a hierarchy of modules or entities that correspond with the original top-level C function hierarchy.
 - Vitis/Vivado HLS automatically inlines sub-functions into higher level functions, or the top-level function as needed to improve performance.
 - By default, each call of the C sub-function uses the same instance of the RTL module.
- Loops in the C functions are kept rolled and are pipelined by default to improve performance.
 - The Vitis/Vivado HLS tool will not unroll loops unless it improves the performance of the solution.
 - Loops can be manually unrolled.
 - Loops can also be pipelined, either with a finite-state machine fine-grain implementation (loop pipelining) or with a more coarse-grain handshake-based implementation (dataflow).
- Arrays in the code are synthesized into block RAM (BRAM), LUT RAM, or UltraRAM in the final FPGA design.
 - If the array is on the top-level function interface, high-level synthesis implements the array as ports with access to a block RAM outside the design.
 - The type of memory used can be reconfigured, or read/write memory transfers can be reconfigured.

After synthesis, you can analyze the results in the various reports produced to determine the quality of your results. After analyzing the results, you can create additional solutions for your project

specifying different constraints and optimization directives, and synthesize and analyze those results. You can compare results between different solutions to see what has worked and what has not. You can repeat this process until the design has the desired performance characteristics. Using multiple solutions allows you to proceed with development while retaining prior solutions.

These are the performance metrics:

- **Area:** Amount of hardware resources required to implement the design based on the resources available in the FPGA, including look-up tables (LUT), registers, block RAMs, and DSP48s.
- **Latency:** Number of clock cycles required for the function to compute all output values.
- **Initiation interval (II):** Number of clock cycles before the function can accept new input data.
- **Loop iteration latency:** Number of clock cycles it takes to complete one iteration of the loop.
- **Loop initiation interval:** Number of clock cycles before the next iteration of the loop starts to process data.
- **Loop latency:** Number of cycles to execute all iterations of the loop.

14 Creating project with Vitis HLS

Inputs of the Vitis/Vivado HLS are:

- **C function written in C, C++11, C++14, or SystemC**
is the primary input to Vitis/Vivado HLS. The function can contain a hierarchy of subfunctions.
- **Constraints**
are required and include the clock period, clock uncertainty, and FPGA target. The clock uncertainty defaults to 12.5% of the clock period if not specified.
- **Directives**
are optional and direct the synthesis process to implement a specific behavior or optimization.
- **C test bench and any associated files**
are needed to simulate the C function prior to synthesis and to verify the RTL output using C/RTL Cosimulation.

Outputs of the Vitis/Vivado HLS are:

- **RTL implementation files in hardware description language (HDL) formats**
This is the primary output from Vitis/Vivado HLS. RTL IP produced by Vitis/Vivado HLS is available in both Verilog (IEEE 1364-2001), and VHDL (IEEE 1076-2000) standards, and can be synthesized and implemented into Xilinx devices.
- **Report files**
This output is the result of synthesis, C/RTL co-simulation, and IP packaging.
- **Compiled object files (.xo)(Vitis)**
This output lets you create compiled hardware functions for use in the Vitis application acceleration development flow. Vitis/Vivado HLS produces this output when called as part of the compilation process from the Vitis tool flow, or when invoked as a stand-alone tool in the bottom up flow.

The shows an overview of the Vitis/Vivado HLS input and output files.

Vitis/Vivado HLS Design Flow

Figure 10: Vitis/Vivado HLS Design Flow

14.1 Coding C/C++ Functions

14.1.1 Coding Style

In any C program, the top-level function is called `main()`. In the Vitis HLS design flow, you can specify any sub-function below `main()` as the top-level function for synthesis. You cannot synthesize the top-level function `main()`. Following are additional rules:

- Only one function is allowed as the top-level function for synthesis.
- Any sub-functions in the hierarchy under the top-level function for synthesis are also synthesized.
- If you want to synthesize functions that are not in the hierarchy under the top-level function for synthesis, you must merge the functions into a single top-level function for synthesis.

14.1.2 C/C++ Language Support

Vitis HLS supports the C/C++ 11/14 for compilation/simulation. Vivado HLS supports the following standards for C compilation/simulation: ANSI-C (GCC 4.6), C++ (G++ 4.6), SystemC (IEEE 1666-2006, version 2.2).

Vitis/Vivado HLS supports many C and C++ language constructs, and all native data types for each language, including float and double types. However, synthesis is not supported for some constructs, including:

- **Dynamic memory allocation**
An FPGA has a fixed set of resources, and the dynamic creation and freeing of memory resources is not supported.
- **Operating system (OS) operations**
All data to and from the FPGA must be read from the input ports or written to output ports. OS operations, such as file read/write or OS queries like time and date, are not supported. Instead, the host application or test bench can perform these operations and pass the data into the function as function arguments.

14.1.3 Libraries in Vitis HLS

Vitis HLS provides foundational C libraries allowing common hardware design constructs and functions to be easily modeled in C and synthesized to RTL. Vitis HLS provides the following C libraries to extend the standard C languages:

- **Arbitrary Precision Data Types Library:** Arbitrary precision data types let your C code use variables with smaller bit-widths than standard C or C++ data types, to enable improved performance and reduced area in hardware.
- **Vitis HLS Math Library:** Used to specify standard math operations for synthesis into RTL and implementation on Xilinx devices.

- HLS Stream Library: For modeling and compiling streaming data structures.
- In addition, the Vitis accelerated libraries are available for use with Vitis HLS, including common functions of math, statistics, linear algebra and DSP; and also supporting domain specific applications, like vision and image processing, quantitative finance, database, data analytics, and data compression. Documentation for the Vitis accelerated libraries can be found at https://xilinx.github.io/Vitis_Libraries/

14.2 Specifying the Clock Frequency

For C and C++ designs only a single clock is supported. The same clock is applied to all functions in the design. The clock period, can be set in a particular solution. Vitis HLS uses the concept of a clock uncertainty to provide a user defined timing margin. The default clock uncertainty, when it is not specified, is 27% of the clock period.

Using the clock frequency and device target information Vitis HLS estimates the timing of operations in the design but it cannot know the final component placement and net routing: these operations are performed by logic synthesis of the output RTL. As such, Vitis HLS cannot estimate the exact delays.

Vitis HLS aims to satisfy all constraints: timing, throughput, latency. However, if a constraints cannot be satisfied, Vitis HLS always outputs an RTL design.

15 C simulation verification

Verification in the Vitis HLS flow can be separated into two distinct processes.

1. Pre-synthesis validation that the C program correctly implements the required functionality which is referred to as **C simulation**.
2. Post-synthesis verification that the generated RTL code performs as expected which is referred to as **C/RTL co-simulation**.

Using C to develop and validate the algorithm before synthesis is much faster than developing and debugging RTL code. Vitis HLS uses the test bench to compile and execute the C simulation. Vitis HLS also uses the same test bench to verify the RTL output of synthesis during C/RTL Co-Simulation in Vitis HLS. The first step in high-level synthesis should be to validate the C code functionality, before generating RTL code, by performing simulation using a well written test bench.

15.1 Writing a Test Bench

1. A C test bench includes a `main()` as a top-level function, that calls the function to be synthesized by the Vitis HLS project.
2. The test bench includes the `main()` function which verifies the top-level functionality for synthesis by providing stimuli and calling the function for synthesis, and by consuming and validating its output.
3. The test bench should execute the top-level function for multiple transactions, allowing many different data values to be applied and verified.
4. The test bench is only as good as the variety of tests it performs.

5. In addition, the test bench must provide multiple transactions if II is to be calculated during RTL simulation.
6. The test bench should ideally be **Self-checking**, and should validate the results from the function to be synthesized are correct.
7. If the results are correct the test bench returns a value of 0 to main(). Otherwise, the test bench should return any non-zero value.
8. The test bench can return any non-zero value. A complex test bench can return different values depending on the type of failure. If the test bench returns a non-zero value after C simulation or C/RTL co-simulation, Vitis HLS reports an error and simulation fails.
9. As Vitis HLS reuses the C test bench for RTL verification, it requires that the test bench and any associated files be denoted as test bench files when they are added to the Vitis HLS project.

16 C Synthesis

During C synthesis, the C/C++ source code is synthesized into an RTL implementation.

Parameters of the C synthesis:

- Latency: Number of clock cycles required for the function to compute all output values.
- Initiation interval (II): Number of clock cycles before the function can accept new input data.
- Loop iteration latency: Number of clock cycles it takes to complete one iteration of the loop.
- Loop iteration interval: Number of clock cycles before the next iteration of the loop starts to process data.
- Loop latency: Number of cycles to execute all iterations of the loop.
- Resource Utilization: Amount of hardware resources required to implement the design based on the resources available in the FPGA, including look-up tables (LUT), registers, block RAMs, and DSP blocks.

17 Analyzing the Results of Synthesis

After synthesis completes, Vitis HLS automatically creates synthesis reports to help you understand and analyze the performance of the implementation. Examples of these reports include the Synthesis Summary report, Schedule Viewer, Function Call Graph, and Dataflow Viewer.

- **Schedule Viewer:** Shows each operation and control step of the function, and the clock cycle that it executes in.
- **Dataflow Viewer:** Shows the dataflow structure inferred by the tool, inspect the channels (FIFO/PIPO), to let you examine the effect of channel depth on performance.
- **Function Call Graph Viewer:** Displays your full design after C Synthesis or C/RTL Cosimulation to show the throughput of the design in terms of latency and II.

The Vitis HLS tool also provides additional views to expand on the information available for analysis of your design.

- **Module Hierarchy:** Shows the resources and latency contribution for each block in the RTL hierarchy. It also indicates any II or timing violations. In case of timing violations, the hierarchy window will also show the total negative slack observed in a specific module.
- **Performance Profile:** Shows details on the performance of the block currently selected in the Module Hierarchy view. Performance is measured in terms of latency and the initiation interval, and includes details on whether the block was pipelined or not.
- **Resource Profile:** Shows the resources used at the selected level of hierarchy, and shows the control state of the operations used.
- **Properties view:** Shows the properties of the currently selected control step or operation in the Schedule Viewer.

17.1 Schedule Viewer

The Schedule Viewer provides a detailed view of the synthesized RTL, showing each operation and control step of the function, and the clock cycle that it executes in. It helps you to identify any loop dependencies that are preventing parallelism, timing violations, and data dependencies. The Schedule Viewer is displayed by default in the Analysis perspective.

17.2 Function Call Graph Viewer

The new Function Call Graph Viewer, which can be opened from the Flow Navigator, illustrates your full design after C Synthesis or C/RTL Co-simulation. The goal of this viewer is to show the throughput of the design in terms of latency and II. It helps identify the critical path in your design and helps you identify bottlenecks in the design to focus on to improve throughput. It can also show the paths through the design where throughput may be imbalanced leading to FIFO stalls and/or deadlock.

In some cases, the displayed hierarchy of the design might not be the same as your source code as a result of HLS optimizations that convert loops into function pipelines, etc. Functions that are in-lined will no longer be visible in the call graph, as they are no longer separate functions in the synthesized code. If multiple instances of a function are created, each unique instance of the function is shown in the call graph.

The call graph displays functions as rectangular boxes, and loops as oval boxes, each with II, latency, and resource or timing data depending on the specific view. Before C/RTL cosimulation is completed the performance and resource metrics that are shown in the graph are from the C Synthesis phase, and are therefore estimates from the HLS tool.

After co-simulation, actual II and latency numbers are reported along with stalling percentages, and this information is back annotated from data collected during co-simulation.

Heat Map feature can also be used to highlight several metrics of interest:

- II (min, max, avg)
- Latency (min, max, avg)
- Stalling Time Percentage

The heat map uses color coding to highlight problematic modules. Using a color scale of red to green where red indicates the high value of the metric (highest II or highest latency) while green indicates a low value of the metric in question. The colors that are neither red nor green represent the range of values that are in between the highest and lowest values. This helps in quickly identifying the modules that need attention.

17.3 Dataflow Viewer

The DATAFLOW optimization is a dynamic optimization which can only be fully understood after the RTL co-simulation is complete. Due to this fact, the Dataflow viewer lets you see the dataflow structure inferred by the tool, inspect the channels (FIFO/PIPO), and examine the effect of channel depth on performance. Performance data is back-annotated to the Dataflow viewer from the co-simulation results.

One must apply the DATAFLOW pragma or directive to the design for the Dataflow viewer to be populated. Dataflow can be applied to the top-level function, or specify regions of a function, or loops. The Dataflow viewer displays a representation of the dataflow graph structure, showing the different processes and the underlying producer-consumer connections.

Features of the Dataflow viewer include the following:

- Source Code browser.
- Automatic cross-probing from process/channel to source code.
- Filtering of ports and channel types.
- Process and Channel table details the characteristics of the design:
 - Channel Profiling (FIFO sizes etc), enabled from Solution Settings dialog box.
 - Process Read Blocking/Write Blocking/Stalling Time reported after RTL co-simulation.
 - Process Latency and II displayed.
 - Channel type and widths are displayed in the Channel table.
 - Automatic cross-probing from Process and Channel table to the Graph and Source browser.
 - Hover over channel or process to display tooltips with design information.

The Dataflow viewer can help with performance debugging your designs. When the design deadlocks during RTL co-simulation, the GUI will open the Dataflow viewer and highlight the channels and processes involved in the deadlock so you can determine what the cause is.

18 Optimizing the HLS Project

After analysis, designer will most likely need or want to optimize the performance of your function. Even if it is performing well there may be opportunities for improvement. One can add optimization directives directly into the source code as compiler pragmas, using various HLS PRAGMAS, or by using Tcl `set_directive` commands to apply optimization directives in a Tcl script to be used by a solution. In addition to optimization pragmas and directives, Vitis HLS provides a number of configuration settings to let you manage the default results of simulation and synthesis.

18.1 Creating Additional Solutions

The most typical use of Vitis HLS is to create an initial design, analyze the results, and then perform optimizations to meet the desired area and performance goals. This is often an iterative process, requiring multiple steps and multiple optimizations to achieve the desired results. Solutions offer a convenient way to configure the tool, add directives to your function to improve the results, and preserve those results to compare with other solutions.

18.2 Adding Pragmas and Directives

Vitis HLS pragmas and directives let you configure the synthesis results for your code.

- HLS Pragmas are added to the source code to enable the optimization or change in the original source code. Every time the code is synthesized, it is implemented according to the specified pragmas.
- Optimization Directives, or the set `_directive` commands, can be specified as Tcl commands that are associated with a specific solution, or set of solutions. Allowing you to customize the synthesis results for the same code base across different solutions.

In cases where pragmas or directives conflict with other pragmas or directives, the synthesis process returns an error until the conflict is resolved.

18.3 Using Directives in Scripts vs Pragmas in Code

In Vitis HLS, there are two destinations possible:

- Directive File: Vitis HLS inserts the directive as a Tcl command into the file `directives.tcl` in the solution directory.
- Source File: Vitis HLS inserts the directive directly into the C source file as a pragma.

18.3.1 Directives file (Tcl Script)

- Advantage: Each solution has independent directives. If any solution is re-synthesized, only the directives specified in that solution are applied.
- Disadvantage: If the C source files are transferred to a third-party or archived, the `directives.tcl` file must be included. The `directives.tcl` file is required if the results are to be re-created.

18.3.2 Source Code (Pragma)

- Advantage: The optimization directives are embedded into the C source code. Ideal when the C sources files are shipped to a third-party as C IP.
- Disadvantage: If the optimization directives are embedded in the code, they are automatically applied to every solution when re-synthesized.

18.3.3 Applying Directives to the Proper Scope

Although the Vitis HLS GUI lets you apply directives to specific code objects, the directives are added to the scope that contains the object. Optimization directives can be applied to the following objects and scopes:

- **Functions:** When you apply directives to functions, Vitis HLS applies the directive to all objects within the scope of that function. The effect of any directive stops at the next level of the function hierarchy, and does not apply to sub-functions.
- **Interfaces:** Vitis HLS applies the directive to the top-level function, which is the scope that contains the interface.
- **Loops:** Directives apply to all objects within the scope of the loop.
- **Arrays:** Directives are applied to the scope that contains the array.

Directives that include a recursive option, such as the PIPELINE directive, can be applied recursively through the hierarchy.

18.3.4 Applying Optimization Directives to Global Variables

Directives can only be applied to scopes, or to objects within a scope. As such, they cannot be directly applied to global variables which are declared outside the scope of any function. Therefore, directive to a global variable should be assigned manually.

18.3.5 Applying Optimization Directives to Class Objects

Optimization directives can be also applied to objects or scopes defined in a class. The difference is typically that classes are defined in a header file.

18.3.6 Applying Optimization Directives to Templates

To apply optimization directives manually on templates when using Tcl commands, specify the template arguments and class when referring to class methods.

19 C/RTL Co-Simulation in Vitis HLS

The C test bench for the simulation purposes, is reused for C/RTL co-simulation to verify that the RTL is functionally identical to the C source code.

19.1 Automatically Verifying the RTL

C/RTL co-simulation uses a C test bench, running the main() function, to automatically verify the RTL design running in behavioral simulation. The C/RTL verification process consists of three phases:

1. The C simulation is executed and the inputs to the top-level function, or the Design-Under-Test (DUT), are saved as "input vectors."
2. The "input vectors" are used in an RTL simulation using the RTL created by Vitis HLS in Vivado simulator, or a supported third-party HDL simulator. The outputs from the RTL, or results of simulation, are saved as "output vectors."
3. The "output vectors" from the RTL simulation are returned to the main() function of the C test bench to verify the results are correct. The C test bench performs verification of the results, in some cases by comparing to known good results.

The following are requirements of C/RTL co-simulation:

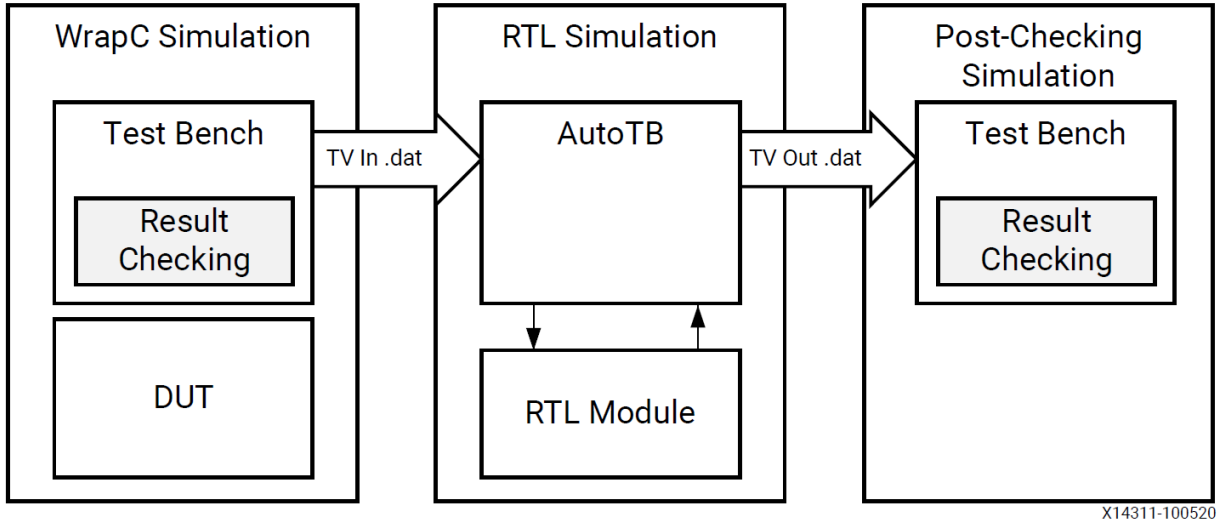


Figure 11: C/RTL Verification Flow

- The test bench must be self-checking as described in Writing a Test Bench, and return a value of 0 if the test passes or returns a non-zero value if the test fails.
- Any third-party simulators must be available in the search path to be launched by Vitis HLS.
- Interface Synthesis Requirements must be met (mentioned below).
- Any arrays or structs on the design interface cannot use the optimization directives listed in Unsupported Optimizations for Co-Simulation.
- IP simulation libraries must be compiled for use with third-party simulators as described in Simulating IP Cores.

Interface Synthesis Requirements: To use the C/RTL co-simulation feature to verify the RTL design, at least one of the following conditions must be true:

- Top-level function must be synthesized using an `ap_ctrl_chain` or `ap_ctrl_hs` blocklevel protocol.
- Design must be purely combinational
- Top-level function must have an initiation interval of 1
- Interfaces must be all arrays that are streaming and implemented with `axis` or `ap_hs` interface modes

19.1.1.1 Verification of DATAFLOW and DEPENDENCE

C/RTL co-simulation automatically verifies aspects of the DATAFLOW and DEPENDENCE directives. If the DATAFLOW directive is used to pipeline tasks, it inserts channels between the tasks to facilitate the flow of data between them. It is typical for the channels to be implemented with FIFOs and the FIFO depth specified using the STREAM directive.

If co-simulation is attempted from the Vitis HLS IDE and the simulation results in a deadlock, the

Vitis HLS IDE will automatically launch the Dataflow Viewer and show the processes involved in the deadlock (displayed in red). It will also show which channels are full (in red) versus empty (in white).

In a similar manner, the RTL test bench is also configured to automatically check the validity of false dependencies specified using the `DEPENDENCE` directive. A warning message during cosimulation indicates the dependency is not false, and the corresponding directive must be removed to achieve a functionally valid design.

19.1.2 Unsupported Optimizations for Co-Simulation

For Vivado IP mode, automatic RTL verification does not support cases where multiple transformations are performed on arrays on the interface, or arrays within structs. In order for automatic verification to be performed, arrays on the function interface, or array inside structs on the function interface, can use any of the following optimizations, but not two or more:

- Vertical mapping on arrays of the same size
- Reshape
- Partition, for dimension 1 of the array

Automatic RTL verification does not support any of the following optimizations used on a top-level function interface:

- Horizontal mapping.
- Vertical mapping of arrays of different sizes.
- Conditional access on the AXI4-Stream with register slice enabled.
- Mapping arrays to streams.

19.2 Analyzing RTL Simulations

When the C/RTL co-simulation completes, the simulation report opens and shows the measured latency and II. These results may differ from values reported after HLS synthesis. The results provided after C/RTL co-simulation show the actual values of latency and II for the given simulation data set (and may change if different input stimuli is used).

In non-pipelined designs, C/RTL co-simulation measures latency between `ap_start` and `ap_done` signals. The II is 1 more than the latency, because the design reads new inputs 1 cycle after all operations are complete.

In pipelined designs, the design might read new inputs before the first transaction completes, and there might be multiple `ap_start` and `ap_ready` signals before a transaction completes. In this case, C/RTL co-simulation measures the latency as the number of cycles between data input values and data output values. The II is the number of cycles between `ap_ready` signals, which the design uses to request new inputs.

19.3 Cosim Deadlock Viewer

A deadlock is a situation in which processes inside a `DATAFLOW` region share the same channels, effectively preventing each other from writing or reading from it, resulting in both processes getting

stuck. This scenario is common when there are either FIFO's or a mix of PIPOs and FIFOs as channels inside the DATAFLOW.

The deadlock viewer visualizes this deadlock scenario on the static dataflow viewer. It highlights the problematic processes and channels. The viewer also provides a cross-probing capability to link between the problematic dataflow channels and the associated source code. The viewer automatically opens only after, the co-simulation detects the deadlock situation and the co-sim run has finished.

19.4 Debugging C/RTL Co-Simulation

When C/RTL co-simulation completes, Vitis HLS typically indicates that the simulations passed and the functionality of the RTL design matches the initial C code.

Following are the primary reasons for a C/RTL co-simulation failure:

- Incorrect environment setup
- Unsupported or incorrectly applied optimization directives
- Issues with the C test bench or the C source code

20 Exporting the RTL Design

The final step in the Vitis HLS flow is to export the RTL design in a form that can be used by other tools in the Xilinx design flow. When Vitis HLS reports the results of the high-level synthesis, it only provides an estimate of the results with projected clock frequencies and resource utilization (LUTs, DSPs, BRAMs, etc.). These results are only estimates because Vitis HLS cannot know what optimizations or routing delays will be in the final synthesized or implemented design. To get a better view of the RTL design, you can actually run Vivado synthesis and place and route on the generated RTL design, and review actual results of timing and resource utilization.

21 Vitis HLS Command Line Interface

Vitis HLS can be run from the GUI, interactively from the command line, or in batch mode from a Tcl script.

21.1 Running Vitis HLS Interactively

You can launch Vitis HLS using the `-i` option to open the tool in interactive mode.

```
vitis_hls -i // Launch Vitis HLS in interactive mode

// The tool then displays a command line prompt
vitis_hls> help // get a list of commands & Help for any command

// Vitis HLS supports an auto-complete feature by pressing the tab key
exit // exit Vitis HLS.
quit // quit Vitis HLS.
```

21.2 Running Vitis HLS in Batch Mode

Vitis HLS can also be run in batch mode, by specifying a Tcl script for the tool to run.

```
vitis_hls -f tcl_script.tcl // Commands embedded in the specified
// script are executed in the specified sequence
```

If the Tcl script includes the exit or quit command, then the tool exits at that point, completing the batch process. If the Tcl script does not end with the exit command, Vitis HLS returns to the command prompt, letting you continue in interactive mode. All of the Tcl commands used when creating a project in the GUI are written to the solution/ script.tcl file within the project. You can use this script as a starting point for developing your own batch scripts. An example script is provided below:

```
open_project dct
set_top dct
add_files ../dct_src/dct.cpp
add_files -tb ../dct_src/out.golden.dat -cflags "-Wno-unknown-pragmas" -
csimflags "-Wno-unknown-pragmas"
add_files -tb ../dct_src/in.dat -cflags "-Wno-unknown-pragmas" -csimflags "-Wno-u
add_files -tb ../dct_src/dct_test.cpp -cflags "-Wno-unknown-pragmas" -
csimflags "-Wno-unknown-pragmas"
open_solution "solution1" -flow_target vitis
set_part {xcvu11p-flga2577-1-e}
create_clock -period 10 -name default
source "../dct/solution1/directives.tcl"
csim_design
csynth_design
cosim_design
export_design -format ip_catalog
```

22 Designing Efficient Kernels

Revisit

23 Vitis HLS Coding Styles

This chapter explains how various constructs of C and C++11/C++14 are synthesized into an FPGA hardware implementation, and discusses any restrictions with regard to standard C coding.

The coding examples in this guide are available on GitHub for use with the Vitis HLS release. You can clone the examples repository from GitHub by clicking the Clone Examples command from the Vitis HLS Welcome screen. Note: To view the Welcome screen at any time, select Help → Welcome.

23.1 Unsupported C/C++ Constructs

While Vitis HLS supports a wide range of the C/C++ languages, some constructs are not synthesizable, or can result in errors further down the design flow. To be synthesized:

- The function must contain the entire functionality of the design.
- None of the functionality can be performed by system calls to the operating system.
- The C/C++ constructs must be of a fixed or bounded size.
- The implementation of those constructs must be unambiguous.

23.1.1 System Calls

System calls cannot be synthesized because they are actions that relate to performing some task upon the operating system in which the C/C++ program is running. Vitis HLS ignores commonly-used system calls that display only data and that have no impact on the execution of the algorithm, such as `printf()` and `fprintf(stdout,)`. In general, calls to the system cannot be synthesized and should be removed from the function before synthesis. Other examples of such calls are `getc()`, `time()`, `sleep()`, all of which make calls to the operating system.

Vitis HLS defines the macro `__SYNTHESIS__` when synthesis is performed. This allows the `__SYNTHESIS__` macro to exclude non-synthesizable code from the design.

Only the `__SYNTHESIS__` macro should be used in the code to be synthesized. Macro in the test bench should not be used, as it is not obeyed by C/C++ simulation or C/C++ RTL co-simulation.

The `__SYNTHESIS__` macro must not be defined or undefined in code or with compiler options, otherwise compilation might fail. The `__SYNTHESIS__` macro is a convenient way to exclude non-synthesizable code without removing the code itself from the function, it can result in different results between C/C++ simulation and C/C++ synthesis.

23.1.2 Dynamic Memory Usage

Any system calls that manage memory allocation within the system, for example, `malloc()`, `alloc()`, and `free()`, are not supported by synthesis. Memory allocation system calls must be removed from the design code before synthesis. Xilinx recommends that you perform the following steps:

1. Add the user-defined macro `NO_SYNTH` to the code and modify the code.
2. Enable macro `NO_SYNTH`, execute the C/C++ simulation, and save the results.
3. Disable the macro `NO_SYNTH`, and execute the C/C++ simulation to verify that the results are identical.
4. Perform synthesis with the user-defined macro disabled.

This methodology ensures that the updated code is validated with C/C++ simulation and that the identical code is then synthesized.

23.1.3 Pointer Limitations

- General Pointer Casting: Vitis HLS does not support general pointer casting, but supports pointer casting between native C/C++ types.
- Pointer Arrays: Vitis HLS supports pointer arrays for synthesis, provided that each pointer points to a scalar or an array of scalars. Arrays of pointers cannot point to additional pointers.
- Function Pointers: Function pointers are not supported.

- Pointer to pointer is not supported.

23.1.4 Recursive Functions

- Recursive functions cannot be synthesized. This applies to functions that can form endless recursion.
- Vitis HLS also does not support tail recursion, in which there is a finite number of function calls.
- Virtual Functions are not supported.

23.1.4.1 Standard Template Libraries Many of the C++ Standard Template Libraries (STLs) contain function recursion and use dynamic memory allocation. For this reason, the STLs cannot be synthesized by Vitis HLS. The solution for STLs is to create a local function with identical functionality that does not feature recursion, dynamic memory allocation, or the dynamic creation and destruction of objects.

23.2 Functions

The top-level function becomes the top level of the RTL design after synthesis. Sub-functions are synthesized into blocks in the RTL design.

The top-level function cannot be a static function.

23.2.1 Inlining Functions

Sub-functions can optionally be inlined to merge their logic with the logic of the surrounding function. While inlining functions can result in better optimizations, it can also increase runtime as more logic must be kept in memory and analyzed.

Vitis HLS can perform automatic inlining of small functions.

If a function is inlined, there is no report or separate RTL file for that function. The logic and loops of the sub-function are merged with the higher-level function in the hierarchy.

23.2.2 C/C++ Builtin Functions

Vitis HLS supports the following C/C++ builtin functions:

- `__builtin_clz(unsigned int x)`: Returns the number of leading 0-bits in x, starting at the most significant bit position. If x is 0, the result is undefined.
- `__builtin_ctz(unsigned int x)`: Returns the number of trailing 0-bits in x, starting at the least significant bit position. If x is 0, the result is undefined.

23.3 Loops

Loops provide a very intuitive and concise way of capturing the behavior of an algorithm and are used often in C/C++ code. Loops are very well supported by synthesis. Loops can be pipelined, unrolled, partially unrolled, merged, and flattened.

The optimizations unroll, partially unroll, flatten, and merge effectively make changes to the loop structure as if the code were changed. These optimizations ensure that limited coding changes are required while optimizing the loops.

Avoid use of global variables for loop index variables, as this can inhibit some optimizations.

The loop implementation techniques are as follows:

1. Rolled implementation / iterative execution
2. Full loop unroll
3. Partial loop unroll
4. Loop pipelining

23.3.1 Rolled implementation

Rolled implementation or iterative execution is a default implementation. In rolled implementation, every iteration will execute in a serial and iterative manner. The loop will execute in ‘n*k’ cycles, where ‘n’ is number of loop iterations and ‘k’ is number of cycles required for each iteration.

- This type of implementation takes maximum time and minimum resources.
- The loops imply latency. Incrementing a loop counter always consumes at least one clock cycle.
- These rolled loops are treated as a single entity; all operations in the loop are implemented using the same hardware resources for each iteration of the loop.

23.3.2 Full loop unroll

In full loop unroll implementation, all the loops are removed by rewriting the operations of iterations of the loop. Loop becomes a basic block and schedule normally. This can be achieved using the following assumption:

There should not be any inter-iteration dependency, only then a full loop unroll can be achieved.

- If a loop is completely unrolled, all operations will be performed in parallel if data dependencies and resources allow.
- In the fully unrolled version all loop operations can be performed in a single clock cycle.
- This type of implementation takes minimum time and maximum resources.
- Loops can be unrolled if their indices are statically determinable at elaboration time. They cannot be unrolled when the number of iterations is variable.

Syntax: `#pragma HLS UNROLL`

23.3.3 Partial loop unroll

Partial loop unroll implementation is a tradeoff between Rolled implementation & Full loop unroll implementation. In this implementation, the loop is unrolled by a certain factor provided by the designer.

As the loop is unrolled partially by a certain factor, the accesses/fetches from a memory are increased by the same factor. Hence, the array or memory must be split or partitioned.

Syntax: `#pragma HLS UNROLL factor=8`

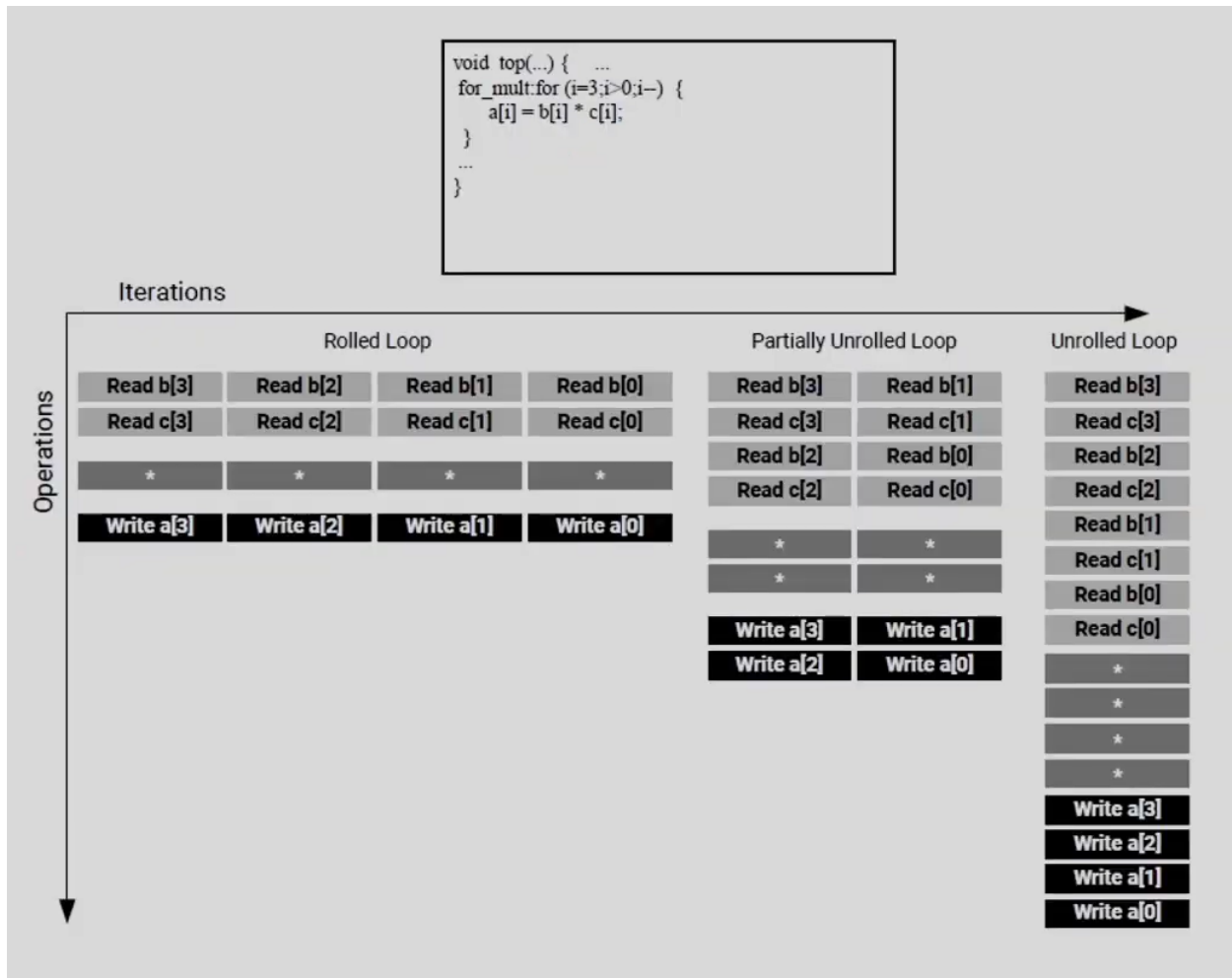


Figure 12: Loop roll, unroll and partially unroll

23.3.4 Loop Pipelining

Loop pipelining implementation allows the operations in a loop to be implemented in an overlapping manner. The pipeline executes until all iterations of the loop are completed.

Syntax: `#pragma HLS PIPELINE II=1`

23.3.5 Nested Loop

Having nested loops in C code is very common. In the case of nested loops:

- Additional clock cycles are required to move between the rolled nested loops.
- One clock cycle is required to move from an outer loop to an inner loop and vice versa.

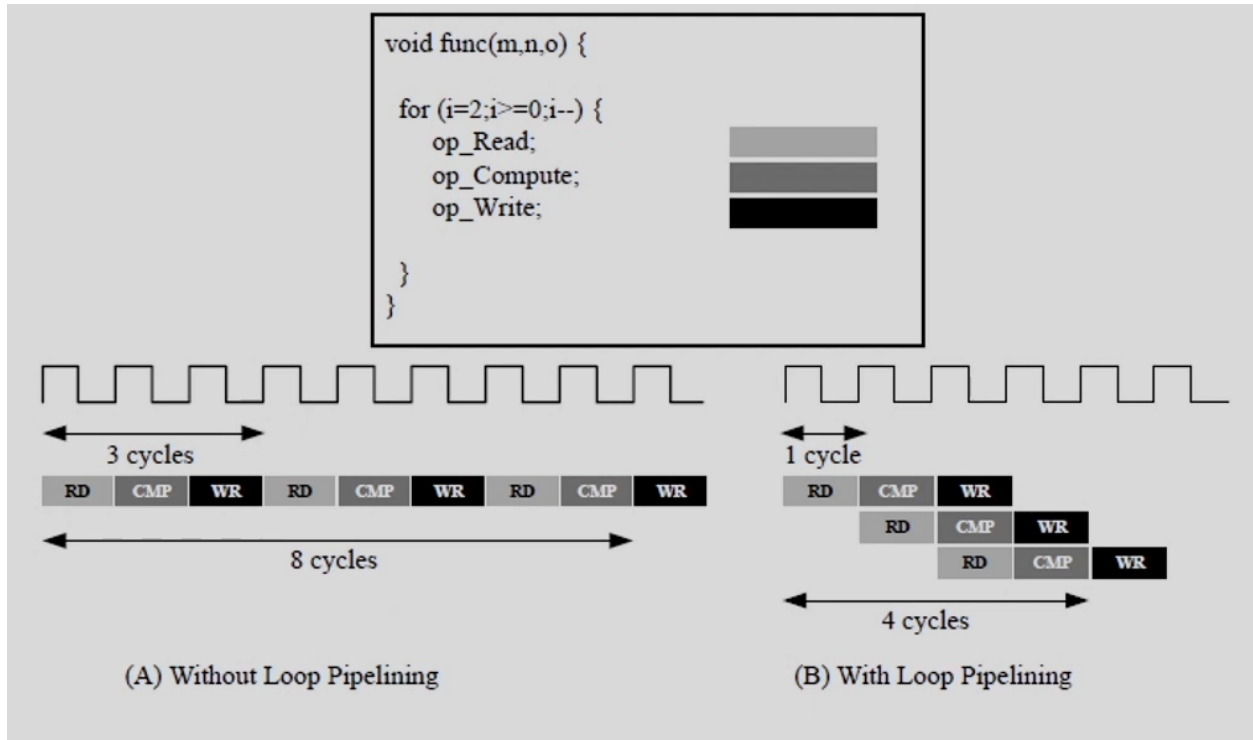


Figure 13: Loop pipelining

- When pipelining loops, the optimal balance between area and performance is typically found by pipelining the innermost loop. This also results in the fastest runtime.
- Pipelining the innermost loop gives the smallest hardware with generally acceptable throughput for most applications.
- When a loop or function is pipelined, any loop in the hierarchy below the loop or function being pipelined must be unrolled.
- There are three cases that can be explored:
 1. Pipelining the innermost loop (Minimum area, Good performance)
 2. Pipelining the outer loop and Unrolling the inner loop (Low area, Better performance)
 3. Pipelining the function (Maximum area, Maximum performance)

The nested loops present in the code can be perfect, semi-perfect, or imperfect loops.

Perfect Loops In a perfect loop, only the innermost loop has the loop body content. There is no logic specified between the loop statements. All the loop bounds are constant.

Semi-Perfect Loops In a semi-perfect loop, only the innermost loop has the loop body content. There is no logic specified between the loop statements. The outermost loop bound can be a variable.

Imperfect Loops For imperfect loop nests where the inner loop has variable bounds or the loop body is not exclusively inside the inner loop, designers should try to restructure the code or unroll the loops in the loop body to create a perfect loop nest. The trivial transformation

from an imperfect to perfect loop (initialization and final write) is made automatically by the tool. When the inner loop of a loop hierarchy is pipelined, Vitis HLS flattens the nested loops to reduce latency and improve overall throughput by removing any cycles caused by loop transitioning.

23.3.6 Loop flattening

The Vitis HLS tool provides the `FLATTEN` directive to allow the nested loops to be flattened, removing the need to recode for optimal hardware performance and reducing the number of cycles it takes to perform the operations in the loop.

The Vitis HLS tool can automatically flatten the nested loops. The recommendation is that the flattening should be specified on the innermost loop. The `off` option can prevent loops in the hierarchy from being flattened.

23.3.7 Loop Merging

Merging the loops allow the logic within the loops to be optimized together. It allows for more efficient architecture explorations.

All rolled loops imply and create at least one state in the design FSM. When there are multiple sequential loops, it can create additional unnecessary clock cycles and prevent further optimizations. The Vitis HLS tool provides the `LOOP_MERGE` optimization directive, which can be used to automatically merge the loops. The `LOOP_MERGE` directive will seek to merge all the loops within the scope it is placed.

23.3.7.1 Loop Merging Rules Currently, the loop merging in the Vitis HLS tool has a few restrictions.

- If loop bounds are all variables, they must have the same value.
- If loops bounds are constants, the maximum constant value is used as the bound of the merged loop.
- Loops with both variable bound and constant bound cannot be merged.
- Code between loops to be merged cannot have side effects: multiple execution of this code should generate the same results.
- Loops cannot be merged when they contain FIFO accesses. Merging would change the order of the reads and writes from a FIFO, which will affect the functioning of the FIFO.
- These loops can be merged using the `force` option.

23.3.8 Variable Loop Bounds

Some of the optimizations that Vitis HLS can apply are prevented when the loop has variable bounds. In this case, Vitis HLS cannot know when the loop will complete. Issues created by variable loop bounds are:

- Variable loop bounds prevent Vitis HLS from determining the latency of the loop as Vitis HLS cannot statically determine the exact value of variable width, and hence, does not know

the number of cycles to completely execute every iteration of the loop. Hence, variable bound loops cannot be unrolled.

- Another issue with variable loop bounds is that the performance of the design is unknown. There are two ways to overcome this issue:
 - Use the pragma HLS loop_tripcount or set_directive_loop_tripcount. The tripcount is the number of loop iterations. The tripcount directive allows a minimum and/or maximum tripcount to be specified for the loop.
 - Use an assert macro in the C/C++ code.

The solution to loops with variable bounds is to make the number of loop iteration a fixed value with conditional executions inside the loop.

```
#define N 32
LOOP_X: for (x=0; x<N; x++) {
    if (x<width) {
        out_accum += A[x];
    }
}
```

23.3.9 Loop Parallelism

Loop Parallelism refers to executing multiple loops parallelly on the FPGA. Vitis HLS schedules logic and functions early as possible to reduce latency while keeping the estimated clock period below the user-specified period. To perform this, it schedules as many logic operations and functions as possible in parallel.

- Vitis HLS does not schedule loops to execute in parallel.
- If the loops have different bounds with same functionality, they cannot be merged. But, by placing the loops in separate functions, the identical functionality can be achieved and both loops can be scheduled in parallel as well.
- The principle of capturing loops in functions to exploit parallelism is presented here for cases in which dataflow optimization cannot be used.
- The dataflow optimization could also be used in the sequential loops.

23.3.10 Loop Dependencies

Loop dependencies are data dependencies that prevent optimization of loops, typically pipelining. They can be within a single iteration of a loop and or between different iteration of a loop. Loop dependencies can occur with any and all types of data. They are particularly common when using arrays.

23.4 Arrays

- Array is a contiguous memory.
- Arrays are usually implemented as memory RAM, ROM , or Registers.

- Arrays on the top-level function interface are synthesized as RTL ports that access a memory outside.
- Internal to the design, arrays sized less than 1024 will be synthesized as FIFO. Arrays sized greater than 1024 will be synthesized into block RAM, LUTRAM, and UltraRAM depending on the optimization settings.
- Array access may create a bottleneck to performance.
- These memories are accessed through ports. There are only a fixed number of ports (typically one or two)

23.4.1 Minimize the parallel array access

To mitigate this bottleneck, the designer has to minimize the parallel array access. It can be done in two ways:

1. Rewriting code to reduce memory access.
2. Using HLS tool features to reduce the number of arrays.

23.4.1.1 Rewriting code to reduce memory access

- Read array in minimum time
- Store locally and reuse
- Reduce memory access as much as possible

23.4.2 Array merging

- For multiple small arrays, there will be wastage of memory if we map each array to a RAM.
- Small arrays can be merged into single block RAM.
- These small arrays should preferably have non-overlapping access.
- Merging can be done either horizontally or vertically. Extra locations in the memory will be padded with zeros.
- Syntax: `#pragma HLS ARRAY_MAP variable instance horizontal/vertical`

23.4.3 Array partitioning

- An array to be used will be larger than standard RAM size or it has many accesses.
- An array can be partitioned into multiple smaller arrays if it has non-overlapping accesses.
- Syntax: `#pragma HLS array_partition variable=<name> <type> factor=<int> dim=<int>`
 - Partition types:
 1. Cyclic: cyclic partitioning creates smaller arrays by interleaving elements from the original array.

2. Block: block partitioning creates smaller arrays from blocks of the original array. This effectively splits the array into N equal blocks.
 3. Complete: complete partitioning decomposes the array into individual elements.
- factor: specifies the number of smaller arrays that are to be created.
 - dim: specifies which dimension of a multi-dimensional array to partition.
- Advantages of array partitioning:
 - Results in RTL with multiple small memories instead of one large memory.
 - Effectively increases the amount of read and write ports for storage.
 - Potentially improves the throughput of the design.

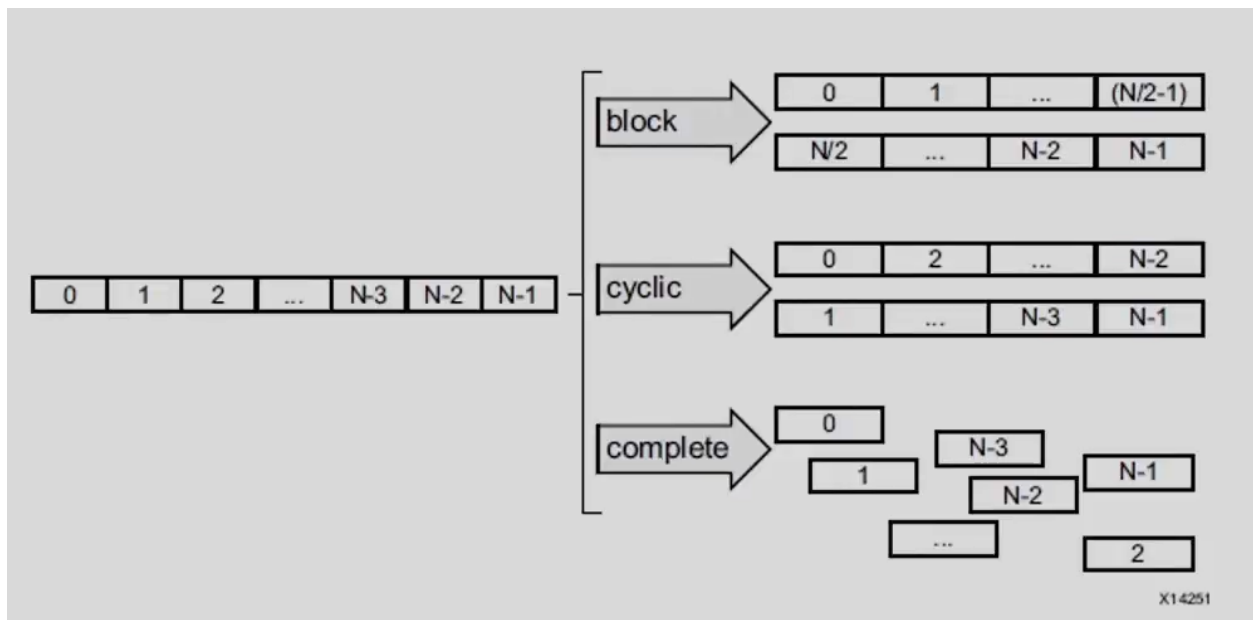


Figure 14: Array partitioning

24 Defining Interfaces

25 Optimization Techniques in Vitis HLS

The following tutorials explain and demonstrate all steps in the process of transforming C, C++ and SystemC code to an RTL implementation using High-Level Synthesis. These show how to create an initial RTL implementation and then transform it into both a low-area and high-throughput implementation by using optimization directives without changing the C code.

Reference for this sections: Vivado Design Suite Tutorial High-Level Synthesis UG871

26 High-Level Synthesis Introduction

This tutorial introduces Vivado High-Level Synthesis (HLS). The primary tasks for performing High-Level Synthesis using both the Graphical User Interface (GUI) and Tcl environments are demonstrated.

26.1 Lab 1: Creating a High-Level Synthesis Project

This lab explains how to set up a High-Level Synthesis (HLS) project and perform all the major steps in the HLS design flow:

1. Creating a New Project:
 1. Open HLS GUI
 2. Enter project details (Name, Location)
 3. Add source files
 4. Add test bench files
 5. Generate Solution (specify hardware details clock period, FPGA part, clock uncertainty)
2. Validate the C code (C Validation or C Simulation):

Aim of this step is to confirm that the C code is correct. This process is also called **C Validation or C Simulation**.
3. High-Level Synthesis:

Synthesize the C design into an RTL design and review the synthesis report.
4. RTL Verification:

High-Level Synthesis can re-use the C test bench to verify the RTL using simulation.
5. IP Creation:

To package the design as an IP block for use with other tools in the Vivado Design Suite.

26.1.1 HLS Graphical User Interface (GUI)

Regions in the Graphical User Interface (GUI) and their functions are:

Vivado HLS Graphical User Interface

Figure 15: Vivado HLS Graphical User Interface

Explorer Pane Shows the project hierarchy.

Information Pane Shows the contents of any files opened from the Explorer pane along with report files.

Auxiliary Pane Cross-links with the Information pane.

Console Pane Shows the messages produced when Vivado HLS runs. Errors and warnings appear in Console pane tabs.

Toolbar Buttons Each button also has an associated menu item available from the pull-down menus to perform the most common operations. When you hold the cursor over the button, a popup tool tip opens, explaining the function.

Perspectives Provide convenient ways to adjust the windows within the Vivado HLS GUI.

Synthesis Perspective The default perspective allows you to synthesize designs, run simulations, and package the IP.

Debug Perspective Includes panes associated with debugging the C code.

Analysis Perspective Windows in this perspective are configured to support analysis of synthesis results.

26.1.2 Usual self check operation of test bench

- The test bench saves the output in a file.
- That file is compared with the expected results stored in file.
- If the output matches the expected results, the return value of the test bench `main()` is set to 0. Else, the return value of `main()` is set to 1.
- If the test bench is self-checking, then the RTL results are automatically checked during RTL verification.
- The Vivado HLS tool can reuse the C test bench to perform verification of the RTL.
- There is no requirement to create an RTL test bench. This provides a robust and productive verification methodology.

26.1.3 C synthesis report

- If the total latency is one clock cycle greater than the loop latency then, this indicates that the design is not pipelined.
- Vivado HLS targets a clock period of Clock Target minus Clock Uncertainty.

Performance Estimates pane Performance Estimates pane enlists clock period, clock uncertainty, latency, Initiation Interval.

Utilization Estimates pane Utilization Estimates pane tries to estimate the resource utilization numbers. These estimations might change post RTL synthesis with additional optimizations.

Interface pane The Interface section shows the ports and I/O protocols created by interface synthesis. HLS automatically adds a clock and reset port along with a few interface ports.

26.1.4 RTL co-simulation

The default option for RTL co-simulation is to perform the simulation using the Vivado simulator and Verilog RTL.

26.2 Lab 2: Using the Tcl Command Interface

This lab exercise shows how to create a Tcl command file based on an existing Vivado HLS project and use the Tcl interface:

1. Open the Vivado HLS Command Prompt
2. Generate a tcl file (For reference: Lab1 script.tcl)

3. Run **"vivado_hls -f run_hls.tcl"** in the the Vivado HLS Command Prompt (within required directory).

26.2.1 Important points to note

- When a Vivado HLS project is created, Tcl files(script.tcl & directives.tcl) are automatically generated in the project hierarchy.
- The file script.tcl contains the Tcl commands to create a project with the files specified during the project setup and run all stages of the HLS flow.
- The file directives.tcl contains any optimizations applied to the design solution.

26.3 Lab 3: Using Solutions for Design Optimization

This lab shows you how to optimize the design using optimization directives. It also creates multiple versions of the RTL implementation and compares the different solutions.

Design goals:

- Create a version of this design with the highest throughput.
- The final design should be able to process data supplied with an input valid signal.
- Produce output data accompanied by an output valid signal.
- The filter coefficients are to be stored externally to the FIR design, in a single port RAM.

1. Creating a New Project.

2. Optimize the I/O Interfaces:

The type of I/O protocol affects the design optimizations possibilities. Add directives from the Auxiliary Pane. Create multiple solutions with different optimizations.

3. Analyze the Results:

Using Analysis perspective, performance and resources can be observed, analyzed.

4. Optimize for the Highest Throughput (Lowest Interval):

Unroll rolled, partition block RAM into individual registers.

5. Compare the results of different solutions.

- If there is an I/O protocol requirement, setting the I/O protocol should be done as early as possible in the design cycle.
- Control states are the internal states High-Level Synthesis uses to schedule operations into clock cycles. There is a close correlation between the control states and the final states in the RTL Finite State Machine (FSM), but there is no one-to-one mapping.
- With the insight gained through analysis, you can proceed to optimize the design.

The two issues that limit the throughput are:

- Rolled loops
- Use of block RAM instead of a shift-register

27 C Validation

Validation of the C algorithm is an important part of the High-Level Synthesis (HLS) process. The time spent ensuring the C algorithm is performing the correct operation and creating a C test bench, which confirms the results are correct, reduces the time spent analyzing designs that are incorrect “by design” and ensures the RTL verification can be performed automatically.

The sample design used in this tutorial is a Hamming Window FIR. There are three versions of this design:

- Using native C data types.
- Using ANSI C arbitrary precision data types.
- Using C++ arbitrary precision data types.

There are no design goals for this tutorial.

27.1 Lab 1: C Validation and Debug

Reviews the aspects of a good C test bench, the basic operations for C validation and the C debugger.

1. Creating a New Project.
2. Review Test Bench and Run C Simulation
3. Run the C Debugger

27.1.1 Good practices for writing a testbench

- The test bench creates and stores a set of expected results that confirm the function is correct.
- The test bench asks the Design Under Test (DUT) to generate and store results.
- The actual and expected results are compared. If the comparison fails, the value of variable `err_cnt` is set to a non-zero value.
- The test bench issues a message to the console if the comparison failed, but more importantly returns the results of the comparison.
- This process of checking the results and returning a value of zero if they are correct automates RTL verification.

27.2 Lab 2: C Validation with ANSI C Arbitrary Precision Types

Uses a design with arbitrary precision C types for C Validation.

1. Creating a New Project
2. Run the C Debugger with Launch Debugger:
Simulation is not completed as arbitrary precision types are not supported in debug mode.
3. Run the C Debugger without Launch Debugger:
Run is successful.

IMPORTANT: When working with arbitrary precision types you can use the Vivado HLS debug environment only with C++ or SystemC. When using arbitrary precision types with ANSI C, the debug environment cannot be used. With ANSI C, you must instead use printf or fprintf statements for debugging.

27.3 Lab 3: C Validation with C++ Arbitrary Precision Types

Uses a design with arbitrary precision C++ types for C Validation.

1. Creating a New Project
2. Run the C Debugger with Launch Debugger:
Debugger is used to observe the arbitrary precision types being populated and utilized.

Arbitrary precision types are a powerful means to create high-performance, bit accurate hardware designs.

28 Interface synthesis

Interface synthesis is the process of adding RTL ports to the C design. In addition to adding the physical ports to the RTL design, interface synthesis includes an associated I/O protocol, allowing the data transfer through the port to be synchronized automatically and optimally with the internal logic.

28.1 Lab 1: Block-Level I/O Protocols

Review the function return and block-level protocols.

1. Creating a New Project
2. Create and Review the Default Block-Level I/O Protocol
3. Modify the Block-Level I/O protocol

28.1.1 Important points to be noted

- A clock and reset (single-bit inputs) have been added to the design by HLS.
- A block-level I/O protocol has been added to control the RTL design.

28.1.2 Block level I/O Protocol

The block-level I/O protocol allows the RTL design to be controlled by additional ports independently of the data I/O ports. This I/O protocol is associated with the function itself and not with any of the data ports. The default block-level I/O protocol is called ap_ctrl_hs (the Control Handshake protocol).

Behavior of the signals for block-level I/O protocol ap_ctrl_hs: Insert table here

- The design does not start operation until ap_start is set to logic 1.
- The design indicates it is no longer idle by setting port ap_idle low.

- Output signal `ap_ready` goes high to indicate the design is ready for new inputs on the next clock.
- Output signal `ap_done` indicates when the design is finished and that the value on output port `ap_return` is valid.
- If `ap_start` is held high, the next transaction will start on the next clock cycle.
- In addition, the RTL cosimulation feature requires a block-level I/O protocol to sequence the test bench and RTL design for cosimulation automatically.
- *Cosim only supports the following ‘ap_ctrl_none’ designs:*
 1. Combinational designs
 2. Pipelined design with task interval of 1
 3. Designs with array streaming or `hls_stream` ports

28.1.3 Types of the block-level interface protocol

There are four types of the block-level interface protocol:

1. `ap_ctrl_none`: No block-level I/O control protocol. When the interface protocol `ap_ctrl_none` is used, no block-level I/O protocols are added to the design. The only ports are those for the clock, reset and the data ports.
2. `ap_ctrl_hs`: The block-level I/O control handshake protocol. This protocol is associated with the function return value (this is true even if the function has no return value specified in the code).
3. `ap_ctrl_chain`: The block-level I/O protocol for control chaining. This I/O protocol is primarily used for chaining pipelined blocks together. In addition to the `ap_ctrl_hs` protocol but with an additional input signal, `ap_continue`, which must be high when `ap_done` is asserted for the next transaction to proceed. This allows downstream blocks to apply back-pressure on the system and halt further processing when they are unable to continue accepting new data.
4. `s_axilite`: Can be applied in addition to `ap_ctrl_hs` or `ap_ctrl_chain` to implement the block-level I/O protocol as an AXI Slave Lite interface in place of separate discrete I/O ports.

28.2 Lab 2: Port I/O Protocols

Understand the default I/O protocol for ports and learn how to select an I/O protocol.

1. Creating a New Project
2. Specify the I/O Protocol for Ports

28.2.1 Important points to be noted

- The code does not have a function return, but instead passes the output of the function through the pointer argument.

- The types of I/O protocol that you can add to C function arguments by interface synthesis depends on the argument type.
- Pass-by-value arguments can be implemented with the following I/O protocols:
 1. `ap_none`: No I/O protocol. This is the default for inputs.
 2. `ap_stable`: No I/O protocol.
 3. `ap_ack`: Implemented with an associated output acknowledge port.
 4. `ap_vld`: Implemented with an associated input valid port.
 5. `ap_hs`: Implemented with both input valid and output acknowledge ports.
- Pass-by-reference arguments that can be implemented with the following I/O protocols:
 1. `ap_none`: No I/O protocol. This is the default for inputs.
 2. `ap_stable`: No I/O protocol.
 3. `ap_ack`: Implemented with an associated input acknowledge port.
 4. `ap_vld`: Implemented with an associated output valid port. This is the default for outputs.
 5. `ap_ovld`: Implemented with an associated output valid port (no valid port for the input part of any inout ports).
 6. `ap_hs`: Implemented with both input valid port and output acknowledge ports
 7. `ap_fifo`: A FIFO interface with associated output write and input FIFO full ports.
 8. `ap_bus`: A Vivado HLS bus interface protocol.

28.3 Lab 3: Implementing Arrays as RTL Interfaces

This lab reviews how array ports are implemented and can be partitioned.

1. Creating a New Project (This design has an input array and an output array.)
2. Synthesize Array Function Arguments to RAM Ports
3. Using Dual-Port RAM and FIFO Interfaces
4. Partitioned RAM and FIFO Array interfaces
5. Fully Partitioned Array Interfaces

28.3.1 Important points to be noted

- Array arguments in the C source are by default synthesized into RTL RAM ports.
- High-Level Synthesis allows you to specify a RAM interface as a single-port or dual-port. If not specified, Vivado HLS automatically analyzes the design and selects the number of ports to maximize the data rate.
- An array argument is implemented using multiple RTL ports, only when the loops are unrolled or when the design is pipelined.

- By using a dual-port RAM interface, input data can be accepted at twice the rate as compared to a single-port RAM interface.

28.4 Lab 4: Implementing AXI4 Interfaces

Create an optimized implementation of the design and add AXI4 interfaces.

1. Creating a New Project (This design has an input array and an output array.)
2. Create an Optimized Design with AXI4-Stream Interfaces:
Varying the partitioning and loop unrolling allowed the optimal balance of area and performance.
3. Implementing an AXI4-Lite Interfaces

When AXI4-Lite interface is added, the IP packaging process creates software driver files to enable an external block, typically a CPU, to control this block (start, stop , set port values, review the interrupt status).

29 Arbitrary Precision Types

C/C++ provided data types are fixed to 8-bit boundaries:

- char (8-bit)
- short (16-bit)
- int (32-bit)
- long long (64-bit)
- float (32-bit)
- double (64-bit)
- Exact width integer types such as `int16_t` (16-bit) and `int32_t` (32-bit)

Using standard C data types for hardware design results in unnecessary hardware costs. Operations can use more LUTs and registers than needed for the required accuracy, and delays might even exceed the clock cycle, requiring more cycles to compute the result.

Vivado High-Level Synthesis (HLS) provides a number of bit accurate or arbitrary precision data-types, allowing you to model variables using any (arbitrary) width.

29.1 Lab 1: Arbitrary Precision

This lab synthesizes the same function used in Lab 1 using arbitrary precision fixed-types highlighting how the same design can be converted to the Vivado HLS `ap_fixed` types, retaining the required accuracy but creating a more optimal hardware implementation.

1. Creating a New Project
2. Review Test Bench and Run C Simulation
3. Synthesize the Design and Review Results

High-Level Synthesis can synthesize floating-point types directly into hardware, provided the operations are standard arithmetic operations (+, -, *, %).

29.2 Lab 2: Arbitrary Precision

This lab synthesizes a design using standard C++ floating-point types and reviews the results.

1. Creating a New Project
2. Review Test Bench and Run C Simulation:
The test bench checks the accuracy of the results by comparing standard C++ floating-point types with HLS Arbitrary Precision types. The results are within a specified range of accuracy.
3. Synthesize the Design and Review Results:
Through use of arbitrary precision types, both the latency and the area have reduced (by 50% and 80% respectively), and the operations in the RTL hardware are no larger than necessary.

Changing data types from standard C types to arbitrary precision types, make sure to reduce the size of the data types. This results in smaller operators, reduced area, and fewer clock cycles to complete.

30 Design Analysis

The general design methodology for creating an RTL implementation from C, C++, or SystemC includes the following tasks:

- Synthesizing the design.
- Reviewing the results of the initial implementation.
- Applying optimization directives to improve performance.

These steps can be repeated until the required performance is achieved. Subsequently, the design can be revisited to improve area.

Vivado High-Level Synthesis (HLS) provides a number of bit accurate or arbitrary precision data-types, allowing you to model variables using any (arbitrary) width.

30.1 Lab 1: Design Optimization

This lab uses the insights from a DCT design analysis to applies optimizations and judges the effectiveness of the optimization.

1. Creating a New Project
2. Review the Source Code and Create the Initial Design
3. Review the Performance Using the Synthesis Report
4. Review the Performance Using the Analysis Perspective
5. Apply Loop Pipelining and Review for Loop Optimization
6. Apply Loop Optimization and Review for Bottlenecks
7. Partition Block RAMs and Analyze Concurrency

8. Partition Block RAMs and Apply Dataflow optimization
9. Optimize the Hierarchy for Dataflow

30.1.1 Important points to be noted

- High-level synthesis might automatically inline small functions to improve the quality of results (QoR). This can be prevented by adding the Inline directive with the -off option to any function being automatically inlined.
- The Analysis perspective consists of five panes:
 - Module Hierarchy Pane: Navigate through the hierarchy.
 - Performance Profile Pane: the performance details for a particular level of hierarchy. Also shows how the operations in a particular block are scheduled into clock cycles.
 - Resource Profile Pane: the resource details for a particular level of hierarchy.
 - Console pane: Comprises of Properties, Outline, Warning, and C Source tab etc.
 - Informtion pane: Shows the contents of any files opened along with report files.
- To improve the initiation interval further from this state, 2 methods can be utilized:
 - Pipeline the loops
 - Pipeline the entire function
- Pipelining the function unrolls all the loops within it, and thus greatly increases the area. If the objective is to get the highest possible performance with no regard for area, this may be the best optimization to perform.
- Pipelining loops transforms the latency from
 $\text{Latency} = \text{iteration latency} * (\text{tripcount} * \text{interval})$ to
 $\text{Latency} = \text{iteration latency} + (\text{tripcount} * \text{interval})$
- Bottlenecks are limitations in the flow of data that can prevent the logic blocks from working at their maximum data rate. Such limitations in the data flow can come from a number of sources.
- Another source of bottlenecks is data dependencies in the original source code. In some cases, these data dependencies are inherent in how the algorithm operates, as when a calculation cannot be performed until an earlier calculation has completed. Sometimes, however, the use of an optimization directive or a minor change to the C code can remove them.
- Approaches for identifying issues in the RTL design:
 - Start with the largest latency of interval in the Module Hierarchy report and navigate down the hierarchy to find the source of any large latency or interval.
 - Click the Resource Profile to examine I/O and memory usage.
 - With the use of graphical viewer, look for patterns in the Performance view which indicate a limitation in data flow.

- The Resource view in analysis perspective shows how the resources in the design are used in different control states.

31 Design Optimization

A crucial part of creating high quality RTL designs using High-Level Synthesis is having the ability to apply optimizations to the C code. High-Level Synthesis always tries to minimize the latency of loops and functions. To achieve this, within the loops and functions, it tries to execute as many operations as possible in parallel. At the level of functions, High-Level Synthesis always tries to execute functions in parallel.

In addition to these automatic optimizations, directives are used to:

- Execute multiple tasks in parallel, for example, multiple executions of the same function or multiple iterations of the same loop. This is pipelining.
- Restructure the physical implementation of arrays (block RAMs), functions, loops and ports to improve the availability of data and help data flow through the design faster.
- Provide information on data dependencies, or lack of them, allowing more optimizations to be performed.

The final optimization technique is to modify the C source code to remove unintended dependencies in the code that may limit the performance of the hardware.

31.1 Lab 1: Optimizing a Matrix Multiplier

31.1.1 Aim

To showcase contrast between the uses of loop and function pipelining to create a design that can process one sample per clock. To analyze loop dependencies and data flow limitations or bottlenecks.

A matrix multiplier is used to design to show how to fully optimize a design heavily based on loops. The design goal is to read one sample per clock cycle using a FIFO interface, while minimizing the area.

1. Create a project
2. Synthesize and Analyze the Design:
By default, the code is implemented without pipelining to set the benchmark.
3. Pipeline the Inner Loop:
It fails to achieve the goal due to Carried dependency (mentioned below.)
4. Pipeline the Outer Loop:
It fails to achieve the goal as 2-cycle BRAM read operations overlap.
5. Reshape the Arrays:
Arrays are reshaped using Array Reshape Directive to achieve 1 clock cycle II.
6. Apply FIFO Interfaces:
It is impossible to use a FIFO interface for data access with the code as written. To use a

FIFO interface, the optimization directives available in Vivado HLS are inadequate as the code currently enforces a certain order of reads and writes.

7. Pipeline the Function:

The design latency decreases further however, the area and resources have increased substantially as all the loops are unrolled.

Important points to be noted:

- To improve the initiation interval further from this state, 2 methods can be utilized:
 - Pipeline the loops
 - Pipeline the entire function
- When pipelining nested loops, pipelining the inner-most loop, might affect the highest as it is ran the most number of times.
- Carried dependency: This is a dependency between an operation in one iteration of a loop and an operation in a different iteration of the same loop.
- High-Level Synthesis automatically applies loop flattening, collapsing the nested loops, removing the loop transitions (essentially creating a single loop with more iterations but overall fewer clock cycles).
- Arrays are implemented as block RAMs and arrays which are arguments to the function are implemented as block RAM ports. As a block RAM can only have a maximum of two ports (for dual-port block RAM), reading more than 2 values in one clock cycle is not possible.
- High-Level Synthesis allows arrays to be partitioned, mapped together and re-shaped.
- High-Level Synthesis can only report one schedule error or warning at a time.
- The default behavior of High-Level Synthesis is to produce a design with the highest performance.
- Pipelining loops allows the loops to remain rolled, thus providing a good means of controlling the area.
- The pipelined function results in the best performance.
- There is a trade off between Performance and Area.

31.2 Lab 2: C Code Optimized for I/O Accesses

This lab shows how modifications to the code can help overcome some performance limitations inherent, but unintended, in the code.

In lab 1, the nature of the C code, which specified multiple accesses to the same addresses, prevented streaming interfaces being applied.

- In a streaming interface, the values must be accessed in sequential order.
- HLS cannot decide to change the specification of the algorithm.

1. Create a project

2. Review the code:

- The directives are specified in the code as pragmas.
- For-loops have been added to cache the row and column reads.
- A temporary variable is used for the accumulation and result is written only when the final result is computed for each value.
- Cache for-loops are automatically unrolled.

3. Synthesize the design and verify the RTL using co-simulation:

Successful synthesis with reading one sample every clock cycle using streaming FIFO interfaces.

32 RTL Verification

The High Level Synthesis tool automates the process of RTL verification and allows you to use RTL verification to generate trace files that show the activity of the waveforms in the RTL design.

RTL verification is often called cosimulation or C/RTL cosimulation; as both C and RTL are used in the verification.

32.1 Lab 1: RTL Verification and the C Test Bench

This lab performs RTL verification steps and understands the importance of the C test bench in verifying the RTL.

1. Create a project
2. Perform RTL Verification:
3. Modify the C test bench

32.2 Lab 2: Viewing Trace Files in Vivado

This lab creates RTL trace files and analyzes them using the Vivado Design Suite. The steps involved in the lab are:

1. Create an RTL Trace File using Vivado Simulator.
2. Perform RTL Verification
3. Modify the C test bench

RTL simulation comprises of three phases.

1. WrapC Simulation: The C test bench is executed to generate input stimuli for the RTL design.
2. RTL Simulation: An RTL test bench with newly generated input stimuli is created and the RTL simulation is then performed.
3. Post-Checking Simulation: Finally, the output from the RTL simulation is re-applied to the C test bench to check the results.

RTL verification issues message SIM-1000 if the RTL verification passed.

32.3 Lab 3: Viewing Trace Files in ModelSim

This lab creates RTL trace files and analyzes them using a third-party RTL simulator.

33 Using HLS IP in IP Integrator

RTL from High-Level Synthesis can be packaged and use it inside IP Integrator.

33.1 Lab 1: Integrate HLS IP with a Xilinx IP Block

This lab completes the steps to generate two HLS blocks for the IP catalog and use them in a design with Xilinx IP, an FFT. The objective is to validate and verify the final design using an RTL test bench.

1. Create Vivado HLS IP Blocks
2. Create a Vivado Design Suite Project
3. Add HLS IP to an IP Repository
4. Create a Block Design for RealFFT
5. Verify the Design

34 Using HLS IP in a Zynq AP SoC Design

A common use of High-Level Synthesis design is to create an accelerator for a CPU – to move code that executes on the CPU into the FPGA programmable logic to improve performance.

34.1 Lab 1: Implement Vivado HLS IP on a Zynq Device

This lab integrates both the High-Level Synthesis IP and the software drivers created by HLS to control the IP in a design implemented on a Zynq device.

1. Create a Vivado HLS IP Block
2. Create a Vivado Zynq Project
3. Add HLS IP to the IP Catalog
4. Creating an IP Integrator Block Design of the System
5. Implementing the System
6. Developing Software and Running it on the Zynq System
7. Modify software to communicate with HLS block

34.2 Lab 2: Streaming Data Between the Zynq CPU and HLS Accelerator Blocks

This lab illustrates a common high performance connection scheme for connecting hardware accelerator blocks that consume data originating in the CPU memory and/or producing data destined for it in a streaming manner.